NETWORK DESIGN AND IMPLEMENTATION PROJECT FOR CEYLON HEAD OFFICE

Diploma in Network Engineering

22.1F

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DECLARATION

The thesis is our original work and has not been submitted previously for a degree at this or any other university/institute.

To my knowledge, it does not contain our material published or written by another person, except as acknowledged in the text

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ABSTRACT

Ceylon, a prominent leader in the retail industry, recognizes the critical role of a robust network infrastructure in enabling seamless operations and superior customer experiences. This abstract provides an overview of a strategic network project tailored to the specific needs of Ceylon's headquarters. The project's focus is on revitalizing the network infrastructure to improve performance, security, and overall efficiency.

The primary objective of this project is to optimize the network at Singer Ceylon's headquarters to align with the organization's growth and technological advancements. Leveraging state-of-the-art networking technologies, such as Cisco's enterprise solutions, we will undertake a comprehensive network upgrade. This upgrade will encompass network switches, routers, access points, and cabling infrastructure to enhance network speed, reliability, and scalability.

Network security is of paramount concern, given the sensitive data and critical operations within Ceylon's headquarters. The project will prioritize the implementation of advanced security measures, including intrusion detection systems, next-generation robust access control policies. These measures aim to safeguard Singer's digital assets, customer information, and business continuity against emerging cyber threats.

The project also includes a meticulous network design phase, ensuring that the infrastructure can accommodate future expansion, new technologies, and remote working capabilities thus fostering enhanced collaboration and productivity.

In conclusion, this strategic network project at Ceylon's headquarters aims to modernize and secure the network infrastructure to meet the demands of the digital era. By leveraging cutting-edge technologies and best practices, we aim to create a resilient and efficient network environment that aligns with Ceylon's vision for excellence in retail and customer satisfaction

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List of Abbreviations

- **PAT** – Port Address Translation
- **DHCP** – Dynamic Host Configuration Protocol
- **STP** – Spanning Tree Protocol
- **VTP** – VLAN Trucking Protocol
- **NAT** – Network Address Translation
- **PAT** – Port Address Translation
- **VLAN** – Virtual LAN
- **HSRP** – Host Standby Router Protocol
- **EIGRP** – Enhanced Interior Gateway Routing Protocol
- **DHCP** – Dynamic Host Configuration Protocol
- **DNS** – Domain Name System
- **PING** – Packet Internet Groper
- **PC** – Personal Computer
- **POE** – Power Over Ethernet
- **ACL** – Access Control List
- **AD** – Active Directory
- **ADDS** – Active Directory Domain Service
- **IP** – Internet Protocol
- **MAC** – Media Access Control
- **ISP** – Internet Service Provider

Chapter 01:

INTRODUCTION

Ceylon (PVT) Limited

Ceylon (PVT) Limited, a subsidiary of the multinational group of company, stands as a prominent and trusted name in the retail and consumer electronics industry in Sri Lanka. With a rich history dating back over a century, Ceylon (PVT) Limited has firmly established itself as a household brand synonymous with quality, innovation, and reliability.

The roots of Ceylon (PVT) Limited trace back to 1877 when the company first set foot in the country. Over the years, it has evolved and diversified its offerings to cater to the evolving needs and aspirations of the Sri Lankan population. Today, Ceylon (PVT) Limited operates as a multifaceted conglomerate, providing a wide array of products and services that touch the lives of millions across the island nation.

Ceylon (PVT) Limited's core business revolves around retail, offering an extensive range of consumer electronics, home appliances, furniture, and sewing machines, among other products. The company has earned the trust of generations of Sri Lankans by consistently delivering high-quality goods that enhance everyday living.

Beyond retail, Ceylon (PVT) Limited has played an active role in fostering community development and engagement. Through various corporate social responsibility initiatives, it has made meaningful contributions to education, healthcare, and social welfare programs, reflecting its commitment to giving back to society.

In addition to its robust retail network, Ceylon (PVT) Limited has embraced digital transformation, leveraging e-commerce platforms to provide customers with convenient and accessible shopping experiences in the digital age. This adaptability to changing market dynamics has allowed Ceylon (PVT) Limited to maintain its leadership position in the industry.

Ceylon (PVT) Limited's success is deeply rooted in its unwavering dedication to customer satisfaction, its innovative product offerings, and its commitment to upholding the highest standards of corporate ethics and responsibility. As Sri Lanka continues to grow and evolve, Ceylon (PVT) Limited remains a trusted partner for consumers, offering the latest in

technology and lifestyle products while contributing to the nation's progress. With a rich heritage and a forward-looking vision, Ceylon (PVT) Limited continues to be a symbol of reliability and excellence in the heart of Sri Lanka.

(Note: This is a fictional introduction created for illustrative purposes.)

Chapter 02 :

BACKGROUND

2.1 Goals and Objectives of Our Network Design

Our network design plays a crucial role in supporting and enhancing our business operations. We have established specific goals and objectives for our network infrastructure to ensure seamless connectivity, data security, and efficient service delivery.

01. Reliability and Availability

We designed this to provide uninterrupted connectivity and to ensure that our systems and services are always available to our customers and employees. This helps us maintain a high level of service.

02. Data Security

We prioritize the security of our customers' information. Our network design incorporates robust security measures, as including firewalls, intrusion detection systems, encryption protocols, and access controls

03. Scalability and Flexibility.

As our company grows, our network must be flexible to meet those needs. We have designed this network to be flexible to these future needs without interruption. This can be flexible to increase customer demand.

04. Performance And Speed

Speed and performance are very important aspects of network connectivity. This allows for a seamless user experience. Therefore, we have designed this network aiming at low latency data transmission.

05. Disaster Recovery and Business Continuity

Our network must be strong enough to effectively deal with unexpected events or disasters. We have used technical methods to minimize downtime as well as protect data and ensure business continuity.

2.2 Importance of Network Infrastructure Implementation

The implementation of a reliable and robust network infrastructure is of utmost Importance to Ceylon (PVT) LTD for several reasons,

1. **Seamless Communication**

Seamless communication is the lifeblood of a company's network infrastructure. It underpins efficiency, collaboration, security and many other aspects of modern business operations. Ensuring that communication within the network is reliable and seamless is essential to maximizing the benefits of a well-implemented network infrastructure.

2. **Enhanced Customer Experience**

An enhanced customer experience is directly tied to the quality and reliability of a company's network infrastructure. A well-implemented network infrastructure supports the seamless delivery of services, personalization, security, and scalability, all of which contribute to higher customer satisfaction and loyalty, ultimately benefiting the company's bottom line.

3. **Data Management and Analytics**

Our network infrastructure supports the collection, storage, and analysis of vast amounts of data related to customer profiles, policies, claims, and industry trends. This data-driven approach enables us to make informed business decisions, optimize our offerings.

4. **Competitive Advantage**

A strong network infrastructure gives us the market advantage we can achieve. It allows us to leverage emerging technologies, such as artificial intelligence, machine learning, and data analytics, to gain insights, streamline processes, and develop innovative products and services that meet evolving customer needs.

Chapter 03:

DESIGN OF SOLUTION

Our network architecture is designed entirely based on customer needs. Network topology design and the physical layer implementations are focused here.

3.1 LOGICAL TOPOLOGY

The most appropriate topology was used to construct the hierarchical network. This hierarchical network consists of the following layers,

- The Core Layer
- The Distribution Layer
- The Access Layer

We selected this because it reduces the complexity and number of network devices. This design is often used in small and medium-sized networks where simplicity and cost effectiveness are prioritized.

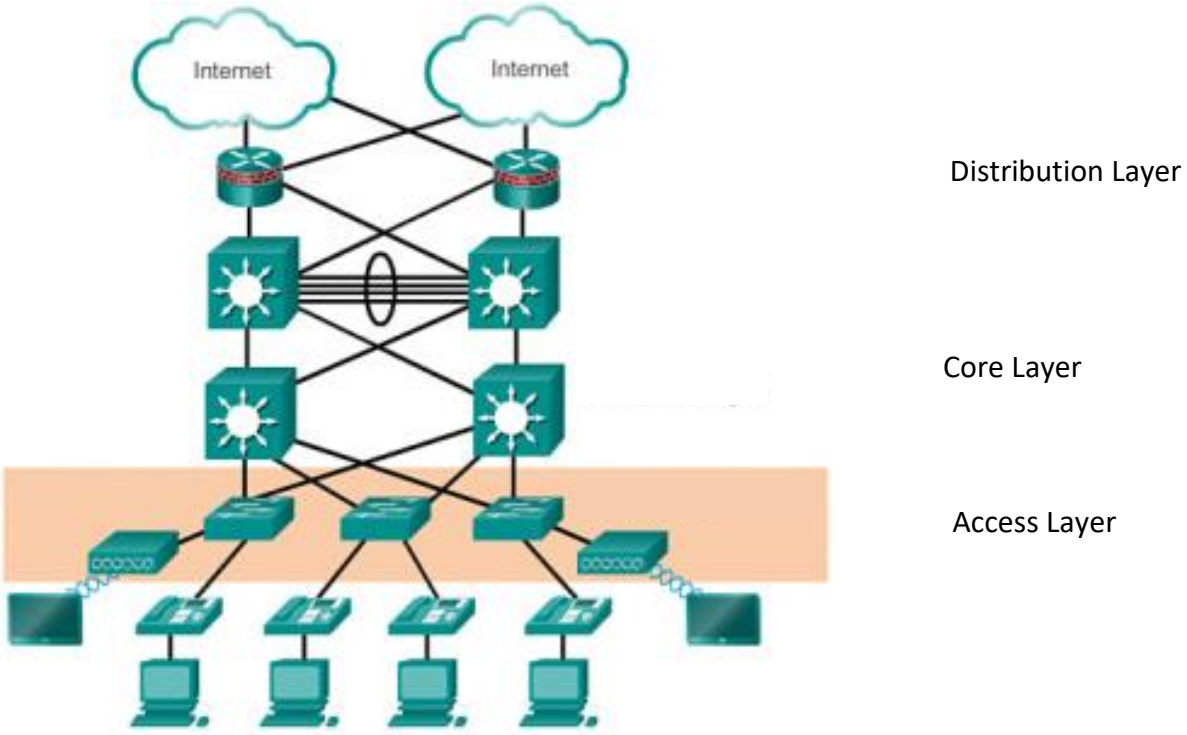


Figure 01 – Hierarchical Network Diagram

3.1.1 TOPOLOGY DIAGRAM

- A collapsed basic network design was implemented for the company's network in accordance with Cisco's hierarchical network design.

- The network mostly uses Cisco equipment because of their well-known reliability and performance. The owner's needs were taken into account when selecting network hardware and protocols.

- Core layer made up of 2-layer 3 support switches. The primary switch (Main switch) is one, and the secondary switch (Backup Switch), which provides redundancy, is the other

- We have used Cat5e cables connect all devices and fiber cables are used to interconnect devices within two stories/sections.

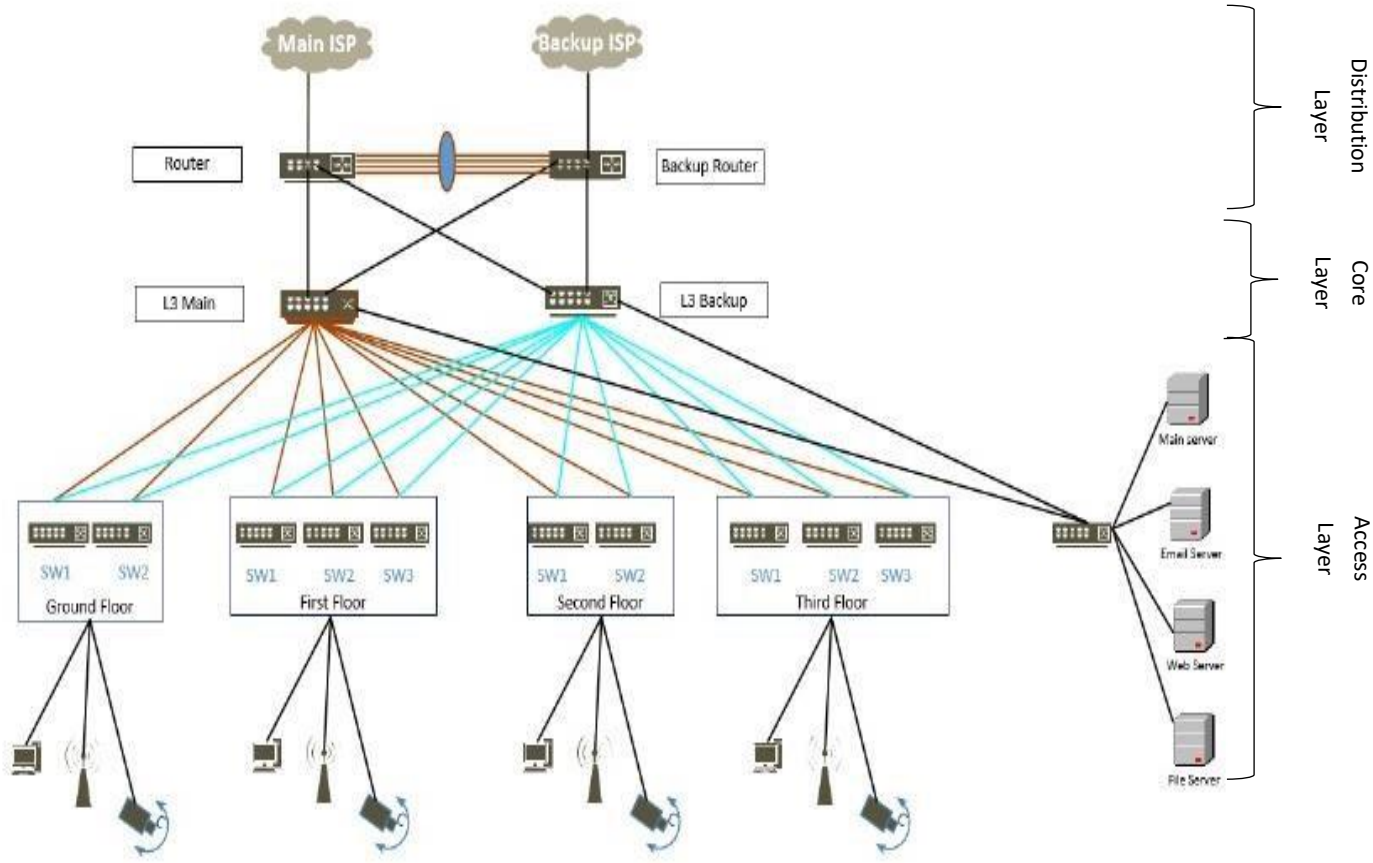


Figure 02 – Network Topology Diagram

- SW – Switch
- L3 Switch – Multilayer Switch
- ISP –Internet Service Provider
- PC – Personal Computer

- Copper cables
- Backup cables (Copper)
- Ether channel

3.1.2 USED TECHNOLOGIES

- **VLAN**

A VLAN is a virtual network created by configuring switches to group devices based on function, location, or other criteria. This improves network efficiency, security, and performance by restricting access and separating traffic.

- **DNS**

DNS is a system that translates domain names into IP addresses for computers to understand, making it easier for users to access websites and online resources. DNS servers store and distribute information about domain names and their corresponding IP addresses, working together in a hierarchical system to handle requests efficiently and accurately.

- **EIGRP**

EIGRP is a Cisco proprietary routing protocol used for exchanging routing information between routers in a network. It uses metrics like bandwidth, delay, and reliability to determine the best path for routing traffic and supports both IPv4 and IPv6 networks. EIGRP is hierarchical, with routers organized into autonomous systems and supports load balancing and route summarization to reduce network traffic and improve performance. It is commonly used in enterprise networks and is considered reliable and efficient.

- **DHCP**

DHCP is a network protocol that automatically assigns IP addresses and other network configuration parameters to devices on a network, eliminating the need for manual configuration. It is typically used in enterprise networks to centrally manage IP address allocation and ensure each device has a unique IP address.

- **PAT**

PAT is a type of NAT that enables multiple devices on a private network to share a single public IP address by assigning unique port numbers to each device. This improves public IP address usage and enhances network security by concealing private network device IP addresses from the public internet. PAT is used in home, small office, and enterprise networks.

- **Ether Channel**

Ether Channel is a networking technology that combines multiple physical links between switches into one logical link, improving network performance and redundancy. It uses

protocols like LACP or PAgP and is commonly used in high-performance environments like data centers and campus networks.

- **STP**

STP is a networking protocol that prevents loops in a network topology by blocking redundant paths between switches, ensuring efficient packet delivery and preventing broadcast storms. It is a standard protocol supported by most network equipment vendors and commonly used in enterprise networks for redundancy and fault tolerance.

- **HSRP**

HSRP is a Cisco protocol that provides redundancy and failover capabilities for IP networks. It allows multiple routers to work together as a virtual router, with one router as the active router and others as standby routers. HSRP uses a virtual IP address as the default gateway for hosts on the network, which remains constant even if the active router fails. It is commonly used in enterprise networks to provide high availability and minimize downtime.

- **SSH**

SSH is a secure network protocol that enables remote access and management of servers and other network devices. It uses encryption and public key cryptography to authenticate users and protect data, making it a reliable and secure means of remote access for system administrators, developers, and IT professionals.

- **ACL**

ACL is an abbreviation for Access Control List, which is a set of rules or permissions used to regulate access to a network or system resource. They are commonly used in computer networks and operating systems to restrict access to files, folders, and other resources based on user identity, group membership, or other criteria. ACLs can be configured to allow or deny access to specific resources, and they help enforce security policies and safeguard sensitive data from unauthorized access.

3.1.3 IP ADDRESSING TABLE

- VLANs in switches provide network segmentation, improved security, and efficient resource utilization.

Device	VLAN no	Network Address	Mask	Subnet Mask	Usable IP Range	Broadcast IP
Wireless Users	100	172.16.0.0	/24	255.255.255.0	172.16.0.1 - 172.16.0.254	172.16.0.255
It Department	40	172.16.1.0	/25	255.255.255.128	172.16.1.1 - 172.16.1.126	172.16.1.127
Training	20	172.16.1.128	/26	255.255.255.192	172.16.1.129 - 172.16.1.190	172.16.1.191
CCTV & Printers	55	172.16.1.192	/26	255.255.255.192	172.16.1.193 - 172.16.1.254	172.16.1.255
Finance Department	30	172.16.2.0	/26	255.255.255.192	172.16.2.1 - 172.16.2.62	172.16.2.63
HR Department	31	172.16.2.64	/26	255.255.255.192	172.16.2.65 - 172.16.2.126	172.16.2.127
Administrative Board	42	172.16.2.128	/27	255.255.255.224	172.16.2.129 - 172.16.2.158	172.16.2.159
Marketing Department	32	172.16.2.160	/27	255.255.255.224	172.16.2.161 - 172.16.2.190	172.16.2.191
Accounts Section	33	172.16.2.192	/27	255.255.255.224	172.16.2.193 - 172.16.2.222	172.16.2.223
Audit	41	172.16.2.224	/27	255.255.255.224	172.16.2.225 - 172.16.2.254	172.16.2.255

Call Centre	13	172.16.3.0	/27	255.255.255.224	172.16.3.1 - 172.16.3.30	172.16.3.31
Director	43	172.16.3.32	/28	255.255.255.240	172.16.3.33 - 172.16.3.46	172.16.3.47
Access Point	50	172.16.3.48	/29	255.255.255.248	172.16.3.49 - 172.16.3.54	172.16.3.55
Inquiry	11	172.16.3.56	/29	255.255.255.248	172.16.3.57 - 172.16.3.62	172.16.3.63
Legal	12	172.16.3.64	/29	255.255.255.248	172.16.3.65 - 172.16.3.70	172.16.3.71
Dining Area	21	172.16.3.72	/29	255.255.255.248	172.16.3.73 - 172.16.3.78	172.16.3.79
Reception	10	172.16.3.80	/30	255.255.255.252	172.16.3.81 - 172.16.3.82	172.16.3.83
Data room	44	172.16.3.84	/30	255.255.255.252	172.16.3.85 - 172.16.3.86	172.16.3.87
Medical Room	22	172.16.3.88	/30	255.255.255.252	172.16.3.89 - 172.16.3.90	172.16.3.91

Table 01 – IP Addressing Table

3.1.4 PORT ASSIGNMENT OF DEVICES

The connection ports of all essential devices are displayed along with their corresponding ports on the switch. All other ports on the switches are inactive except the occupied ports. They will not participate in the data transfer process. To avoid confusion, all ports should be labeled with different labels.

3.1.4.1 Access Layer Switches

1. Ground Floor

Ground Floor – SW_01

INTERFACE	CONNECTS TO	CONNECTING PORTS	VLAN NO	PORT MODE
Fa/0/1	GF-PC-1	F0/1	10	ACCESS
Fa/0/2	GF-PC-2	F0/1	10	ACCESS
Fa/0/3	GF-PC-3	F0/1	11	ACCESS
Fa/0/4	GF-PC-4	F0/1	11	ACCESS
Fa/0/5	GF-PC-5	F0/1	11	ACCESS
Fa/0/6	GF-PC-6	F0/1	11	ACCESS
Fa/0/7	GF-PC-7	F0/1	11	ACCESS
Fa/0/8	GF-PC-8	F0/1	12	ACCESS
Fa/0/9	GF-PC-9	F0/1	12	ACCESS
Fa/0/10	GF-PC-10	F0/1	12	ACCESS
Fa/0/11	GF-PC-11	F0/1	12	ACCESS
Fa/0/12	GF-PC-12	F0/1	12	ACCESS
Fa/0/13	GF-PC-13	F0/1	13	ACCESS
Fa/0/14	GF-PC-14	F0/1	13	ACCESS
Fa/0/15	GF-PC-15	F0/1	13	ACCESS
Fa/0/16	GF-PC-16	F0/1	13	ACCESS
Fa/0/17	GF-PC-17	F0/1	13	ACCESS
Fa/0/18	GF-PC-18	F0/1	14	ACCESS

Fa/0/19	GF-PC-19	F0/1	14	ACCESS
Fa/0/20	GF-PC-20	F0/1	14	ACCESS
Fa/0/21	GF-PC-21	F0/1	14	ACCESS
Fa/0/22	GF-PC-22	F0/1	14	ACCESS
Fa/0/23	GF-PC-23	F0/1	14	ACCESS
Fa/0/24	GF-PC-24	F0/1	14	ACCESS
Fa/0/25	GF-PC-25	F0/1	14	ACCESS
Fa/0/26	GF-PC-26	F0/1	14	ACCESS
Fa/0/27	GF-PC-27	F0/1	14	ACCESS
Fa/0/28	GF-PC-28	F0/1	14	ACCESS
Fa/0/29	GF-PC-29	F0/1	14	ACCESS
Fa/0/30	GF-PC-30	F0/1	14	ACCESS
Fa/0/31	GF-PC-31	F0/1	14	ACCESS
Fa/0/32	GF-PC-32	F0/1	14	ACCESS
Fa/0/33	GF-PC-33	F0/1	14	ACCESS
Fa/0/34	GF-PC-34	F0/1	14	ACCESS
Fa/0/35	GF-PC-35	F0/1	14	ACCESS
Fa/0/36	GF-PC-36	F0/1	14	ACCESS
Fa/0/37	GF-PC-37	F0/1	14	ACCESS
Fa/0/38	GF-AP-1	F0/1	50	ACCESS
Fa/0/39	GF-AP-2	F0/1	50	ACCESS
Fa/0/40	GF-AP-3	F0/1	50	ACCESS
Fa/0/41	GF-AP-4	F0/1	50	ACCESS
Fa/0/42	GF-AP-5	F0/1	50	ACCESS
Fa/0/43	GF-AP-6	F0/1	50	ACCESS
Fa/0/44	GF-AP-7	F0/1	50	ACCESS
Fa/0/45	GF-AP-8	F0/1	50	ACCESS
GigabitEthernet 0/1	CO-SW-1	GigabitEthernet 0/3	-	TRUNK
GigabitEthernet 0/2	CO-SW-2	GigabitEthernet 0/3	-	TRUNK

Table 02 – Ground Floor – SW_01

2. First Floor

First Floor- SW_01

INTERFACE	CONNECTS TO	CONNECTING PORTS	VLAN NO	PORT MODE
Fa/0/1	1F-PC-1	F0/1	20	ACCESS
Fa/0/2	1F-PC-2	F0/1	20	ACCESS
Fa/0/3	1F-PC-3	F0/1	20	ACCESS
Fa/0/4	1F-PC-4	F0/1	20	ACCESS
Fa/0/5	1F-PC-5	F0/1	20	ACCESS
Fa/0/6	1F-PC-6	F0/1	20	ACCESS
Fa/0/7	1F-PC-7	F0/1	20	ACCESS
Fa/0/8	1F-PC-8	F0/1	20	ACCESS
Fa/0/9	1F-PC-9	F0/1	20	ACCESS
Fa/0/10	1F-PC-10	F0/1	20	ACCESS
Fa/0/11	1F-PC-11	F0/1	20	ACCESS
Fa/0/12	1F-PC-12	F0/1	20	ACCESS
Fa/0/13	1F-PC-13	F0/1	20	ACCESS
Fa/0/14	1F-PC-14	F0/1	20	ACCESS
Fa/0/15	1F-PC-15	F0/1	20	ACCESS
Fa/0/16	1F-PC-16	F0/1	20	ACCESS
Fa/0/17	1F-PC-17	F0/1	20	ACCESS
Fa/0/18	1F-PC-18	F0/1	20	ACCESS
Fa/0/19	1F-PC-19	F0/1	20	ACCESS
Fa/0/20	1F-PC-20	F0/1	20	ACCESS
Fa/0/21	1F-PC-21	F0/1	20	ACCESS
Fa/0/22	1F-PC-22	F0/1	20	ACCESS
Fa/0/23	1F-PC-23	F0/1	20	ACCESS
Fa/0/24	1F-PC-24	F0/1	20	ACCESS
Fa/0/25	1F-PC-25	F0/1	20	ACCESS

Fa/0/26	1F-PC-26	F0/1	20	ACCESS
Fa/0/27	1F-PC-27	F0/1	20	ACCESS
Fa/0/28	1F-PC-28	F0/1	20	ACCESS
Fa/0/29	1F-PC-29	F0/1	20	ACCESS
Fa/0/30	1F-PC-30	F0/1	20	ACCESS
Fa/0/31	1F-PC-31	F0/1	20	ACCESS
Fa/0/32	1F-PC-32	F0/1	20	ACCESS
Fa/0/33	1F-PC-33	F0/1	20	ACCESS
Fa/0/34	1F-PC-34	F0/1	20	ACCESS
Fa/0/35	1F-PC-35	F0/1	20	ACCESS
Fa/0/36	1F-PC-36	F0/1	20	ACCESS
Fa/0/37	1F-PC-37	F0/1	20	ACCESS
Fa/0/38	1F-PC-38	F0/1	20	ACCESS
Fa/0/39	1F-PC-39	F0/1	20	ACCESS
Fa/0/40	1F-PC-40	F0/1	20	ACCESS
Fa/0/41	1F-PC-41	F0/1	20	ACCESS
Fa/0/42	1F-PC-42	F0/1	20	ACCESS
Fa/0/43	1F-PC-43	F0/1	20	ACCESS
Fa/0/44	1F-PC-44	F0/1	20	ACCESS
Fa/0/45	1F-PC-45	F0/1	20	ACCESS
Fa/0/46	1F-PC-46	F0/1	20	ACCESS
Fa/0/47	1F-PC-47	F0/1	20	ACCESS
Fa/0/48	1F-PC-48	F0/1	20	ACCESS
GigabitEthernet 0/1	CO-SW-1	GigabitEthernet 0/3	-	TRUNK
GigabitEthernet 0/2	CO-SW-2	GigabitEthernet 0/3	-	TRUNK

Table 03 – First Floor – SW_01

First Floor -SW_02

INTERFACE	CONNECTS TO	CONNECTING PORTS	VLAN NO	PORD MODE
Fa/0/1	1F-PC-49	F0/1	20	ACCESS
Fa/0/2	1F-PC-50	F0/1	20	ACCESS
Fa/0/3	1F-PC-51	F0/1	21	ACCESS
Fa/0/4	1F-PC-52	F0/1	21	ACCESS
Fa/0/5	1F-PC-53	F0/1	21	ACCESS
Fa/0/6	1F-PC-54	F0/1	22	ACCESS
Fa/0/7	1F-AP-1	F0/1	45	ACCESS
Fa/0/8	1F-AP-2	F0/1	45	ACCESS
Fa/0/9	1F-AP-3	F0/1	45	ACCESS
Fa/0/10	1F-AP-4	F0/1	45	ACCESS
Fa/0/11	1F-AP-5	F0/1	45	ACCESS
Fa/0/12	1F-AP-6	F0/1	45	ACCESS
FA/0/13-24	SHUTDOWN	SHUTDOWN	-	SHUTDOWN
GigabitEthernet 0/1	CO-SW-1	GigabitEthernet 0/4	-	TRUNK
GigabitEthernet 0/2	CO-SW-2	GigabitEthernet 0/4	-	TRUNK

Table 04 – First Floor – SW_02

3. Second Floor

Second Floor-SW_01

INTERFACE	CONNECTS TO	CONNECTING PORTS	VLAN NO	PORT MODE
Fa/0/1	F2-PC-FD-1	F0/1	30	ACCESS
Fa/0/2	F2-PC-FD-2	F0/1	30	ACCESS
Fa/0/3	F2-PC-FD-3	F0/1	30	ACCESS
Fa/0/4	F2-PC-FD-4	F0/1	30	ACCESS
Fa/0/5	F2-PC-FD-5	F0/1	30	ACCESS
Fa/0/6	F2-PC-FD-6	F0/1	30	ACCESS
Fa/0/7	F2-PC-FD-7	F0/1	30	ACCESS
Fa/0/8	F2-PC-FD-8	F0/1	30	ACCESS
Fa/0/9	F2-PC-FD-9	F0/1	30	ACCESS
Fa/0/10	F2-PC-FD-10	F0/1	30	ACCESS
Fa/0/11	F2-PC-FD-11	F0/1	30	ACCESS
Fa/0/12	F2-PC-FD-12	F0/1	30	ACCESS
Fa/0/13	F2-PC-FD-13	F0/1	30	ACCESS
Fa/0/14	F2-PC-FD-14	F0/1	30	ACCESS
Fa/0/15	F2-PC-FD-15	F0/1	30	ACCESS
Fa/0/16	F2-PC-FD-16	F0/1	30	ACCESS
Fa/0/17	F2-PC-FD-17	F0/1	30	ACCESS
Fa/0/18	F2-PC-FD-18	F0/1	30	ACCESS
Fa/0/19	F2-PC-FD-19	F0/1	30	ACCESS
Fa/0/20	F2-PC-FD-20	F0/1	30	ACCESS
Fa/0/21	F2-PC-FD-21	F0/1	30	ACCESS
Fa/0/22	F2-PC-FD-22	F0/1	30	ACCESS
Fa/0/23	F2-PC-FD-23	F0/1	30	ACCESS
Fa/0/24	F2-PC-FD-24	F0/1	30	ACCESS
Fa/0/25	F2-PC-FD-25	F0/1	30	ACCESS
Fa/0/26	F2-PC-FD-26	F0/1	30	ACCESS

Fa/0/27	F2-PC-FD-27	F0/1	30	ACCESS
Fa/0/28	F2-PC-FD-28	F0/1	30	ACCESS
Fa/0/29	F2-PC-FD-29	F0/1	30	ACCESS
Fa/0/30	F2-PC-FD-30	F0/1	30	ACCESS
Fa/0/31	F2-PC-FD-31	F0/1	30	ACCESS
Fa/0/32	F2-PC-FD-32	F0/1	30	ACCESS
Fa/0/33	F2-PC-FD-33	F0/1	30	ACCESS
Fa/0/34	F2-PC-FD-34	F0/1	30	ACCESS
Fa/0/35	F2-PC-FD-35	F0/1	30	ACCESS
Fa/0/36	F2-PC-HR-1	F0/1	31	ACCESS
Fa/0/37	F2-PC-HR-2	F0/1	31	ACCESS
Fa/0/38	F2-PC-HR-3	F0/1	31	ACCESS
Fa/0/39	F2-PC-HR-4	F0/1	31	ACCESS
Fa/0/40	F2-PC-HR-5	F0/1	31	ACCESS
Fa/0/41	F2-PC-HR-6	F0/1	31	ACCESS
Fa/0/42	F2-PC-HR-7	F0/1	31	ACCESS
Fa/0/43	F2-PC-HR-8	F0/1	31	ACCESS
Fa/0/44	F2-PC-HR-9	F0/1	31	ACCESS
Fa/0/45	F2-PC-HR-10	F0/1	31	ACCESS
Fa/0/46	F2-PC-HR-11	F0/1	31	ACCESS
Fa/0/47	F2-PC-HR-12	F0/1	31	ACCESS
Fa/0/48	F2-PC-HR- 13	F0/1	31	ACCESS
GigabitEthernet 0/1	CO-SW-1	GigabitEthernet 0/7	-	TRUNK
GigabitEthernet 0/2	CO-SW-2	GigabitEthernet 0/7	-	TRUNK

Table 05 – Second Floor – SW_01

Second Floor-SW_02

INTERFACE	CONNECTS TO	CONNECTING PORTS	VLAN NO	PORT MODE
Fa/0/1	F2-PC-HR-14	F0/1	31	ACCESS
Fa/0/2	F2-PC-HR-15	F0/1	31	ACCESS
Fa/0/3	F2-PC-HR-16	F0/1	31	ACCESS
Fa/0/4	F2-PC-HR-17	F0/1	31	ACCESS
Fa/0/5	F2-PC-HR-18	F0/1	31	ACCESS
Fa/0/6	F2-PC-HR-19	F0/1	31	ACCESS
Fa/0/7	F2-PC-HR-20	F0/1	31	ACCESS
Fa/0/8	F2-PC-HR-21	F0/1	31	ACCESS
Fa/0/9	F2-PC-HR-22	F0/1	31	ACCESS
Fa/0/10	F2-PC-HR-23	F0/1	31	ACCESS
Fa/0/11	F2-PC-HR-24	F0/1	31	ACCESS
Fa/0/12	F2-PC-HR-25	F0/1	31	ACCESS
Fa/0/13	F2-PC-HR-26	F0/1	31	ACCESS
Fa/0/14	F2-PC-HR-27	F0/1	31	ACCESS
Fa/0/15	F2-PC-HR-28	F0/1	31	ACCESS
Fa/0/16	F2-PC-HR-29	F0/1	31	ACCESS
Fa/0/17	F2-PC-HR-0	F0/1	31	ACCESS
Fa/0/18	F2-PC-HR-31	F0/1	31	ACCESS
Fa/0/19	F2-PC-HR-32	F0/1	31	ACCESS
Fa/0/20	F2-PC-HR-33	F0/1	31	ACCESS
Fa/0/21	F2-PC-HR-34	F0/1	31	ACCESS
Fa/0/22	F2-PC-HR-35	F0/1	31	ACCESS
Fa/0/23	F2-PC-MD-1	F0/1	32	ACCESS
Fa/0/24	F2-PC-MD-2	F0/1	32	ACCESS
Fa/0/25	F2-PC-MD-3	F0/1	32	ACCESS
Fa/0/26	F2-PC-MD-4	F0/1	32	ACCESS
Fa/0/27	F2-PC-MD-5	F0/1	32	ACCESS

Fa/0/28	F2-PC-MD-6	F0/1	32	ACCESS
Fa/0/29	F2-PC-MD-7	F0/1	32	ACCESS
Fa/0/30	F2-PC-MD-8	F0/1	32	ACCESS
Fa/0/31	F2-PC-MD-9	F0/1	32	ACCESS
Fa/0/32	F2-PC-MD-10	F0/1	32	ACCESS
Fa/0/33	F2-PC-MD-11	F0/1	32	ACCESS
Fa/0/34	F2-PC-MD-12	F0/1	32	ACCESS
Fa/0/35	F2-PC-MD-13	F0/1	32	ACCESS
Fa/0/36	F2-PC-MD-14	F0/1	32	ACCESS
Fa/0/37	F2-PC-MD-15	F0/1	32	ACCESS
Fa/0/38	F2-PC-MD-16	F0/1	32	ACCESS
Fa/0/39	F2-PC-MD-17	F0/1	32	ACCESS
Fa/0/40	F2-PC-MD-18	F0/1	32	ACCESS
Fa/0/41	F2-PC-MD-19	F0/1	32	ACCESS
Fa/0/42	F2-PC-MD-20	F0/1	32	ACCESS
Fa/0/43	F2-PC-MD-21	F0/1	32	ACCESS
Fa/0/44	F2-PC-MD-22	F0/1	32	ACCESS
Fa/0/45	F2-PC-MD-23	F0/1	32	ACCESS
Fa/0/46	F2-PC-MD-24	F0/1	32	ACCESS
Fa/0/47	F2-PC-MD-25	F0/1	32	ACCESS
Fa/0/48	F2-PC-MD-25	F0/1	32	ACCESS
GigabitEthernet 0/1	CO-SW-1	GigabitEthernet 0/3	-	TRUNK
GigabitEthernet 0/2	CO-SW-2	GigabitEthernet 0/3	-	TRUNK

Table 06 – Second Floor – SW_02

Second Floor- SW_03

INTERFACE	CONNECTS TO	CONNECTING PORTS	VLAN NO	PORT MODE
Fa/0/1	F2-PC-MD-27	F0/1	32	ACCESS
Fa/0/2	F2-PC-MD-28	F0/1	32	ACCESS
Fa/0/3	F2-PC-MD-29	F0/1	32	ACCESS
Fa/0/4	F2-PC-AS-1	F0/1	33	ACCESS
Fa/0/5	F2-PC-AS-2	F0/1	33	ACCESS
Fa/0/6	F2-PC-AS-3	F0/1	33	ACCESS
Fa/0/7	F2-PC-AS-4	F0/1	33	ACCESS
Fa/0/8	F2-PC-AS-5	F0/1	33	ACCESS
Fa/0/9	F2-PC-AS-6	F0/1	33	ACCESS
Fa/0/10	F2-PC-AS-7	F0/1	33	ACCESS
Fa/0/11	F2-PC-AS-8	F0/1	33	ACCESS
Fa/0/12	F2-PC-AS-9	F0/1	33	ACCESS
Fa/0/13	F2-PC-AS-10	F0/1	33	ACCESS
Fa/0/14	F2-PC-AS-11	F0/1	33	ACCESS
Fa/0/15	F2-PC-AS-12	F0/1	33	ACCESS
Fa/0/16	F2-PC-AS-13	F0/1	33	ACCESS
Fa/0/17	F2-PC-AS-14	F0/1	33	ACCESS
Fa/0/18	F2-PC-AS-15	F0/1	33	ACCESS
Fa/0/19	F2-PC-AS-16	F0/1	33	ACCESS
Fa/0/20	F2-PC-AS-17	F0/1	33	ACCESS
Fa/0/21	F2-PC-AS-18	F0/1	33	ACCESS
Fa/0/22	F2-PC-AS-19	F0/1	33	ACCESS
Fa/0/23	F2-PC-AS-20	F0/1	33	ACCESS
Fa/0/24	F2-AP-1	F0/1	50	ACCESS
Fa/0/25	F2-AP-2	F0/1	50	ACCESS
Fa/0/26	F2-AP-3	F0/1	50	ACCESS
Fa/0/27	F2-AP-4	F0/1	50	ACCESS

Fa/0/28	F2-AP-5	F0/1	50	ACCESS
GigabitEthernet 0/1	CO-SW-1	GigabitEthernet 0/9	-	TRUNK
GigabitEthernet 0/2	CO-SW-2	GigabitEthernet 0/9	-	TRUNK

Table 07 – Second Floor – SW_03

4.Third Floor

Third Floor- SW_01

INTERFACE	CONNECTS TO	CONNECTING PORTS	VLAN NO	PORT MODE
Fa/0/1	F3-IT-PC-1	F0/1	40	ACCESS
Fa/0/2	F3-IT-PC-2	F0/1	40	ACCESS
Fa/0/3	F3-IT -PC-3	F0/1	40	ACCESS
Fa/0/4	F3-IT -PC-4	F0/1	40	ACCESS
Fa/0/5	F3-IT -PC-5	F0/1	40	ACCESS
Fa/0/6	F3-IT -PC-6	F0/1	40	ACCESS
Fa/0/7	F3-IT -PC-7	F0/1	40	ACCESS
Fa/0/8	F3-IT-PC-8	F0/1	40	ACCESS
Fa/0/9	F3-IT -PC-9	F0/1	40	ACCESS
Fa/0/10	F3-IT -PC-10	F0/1	40	ACCESS
Fa/0/11	F3-IT -PC-11	F0/1	40	ACCESS
Fa/0/12	F3-IT -PC-12	F0/1	40	ACCESS
Fa/0/13	F3-IT -PC-13	F0/1	40	ACCESS
Fa/0/14	F3-IT -PC-14	F0/1	40	ACCESS
Fa/0/15	F3-IT -PC-15	F0/1	40	ACCESS
Fa/0/16	F3-IT -PC-16	F0/1	40	ACCESS
Fa/0/17	F3-IT -PC-17	F0/1	40	ACCESS
Fa/0/18	F3-IT -PC-18	F0/1	40	ACCESS
Fa/0/19	F3-IT -PC-19	F0/1	40	ACCESS
Fa/0/20	F3-IT-PC-20	F0/1	40	ACCESS
Fa/0/21	F3-IT -PC-21	F0/1	40	ACCESS

Fa/0/22	F3-IT -PC-22	F0/1	40	ACCESS
Fa/0/23	F3-IT-PC-23	F0/1	40	ACCESS
Fa/0/24	F3-IT -PC-24	F0/1	40	ACCESS
Fa/0/25	F3-IT -PC-25	F0/1	40	ACCESS
Fa/0/26	F3-IT -PC-26	F0/1	40	ACCESS
Fa/0/27	F3-IT -PC-27	F0/1	40	ACCESS
Fa/0/28	F3-IT -PC-28	F0/1	40	ACCESS
Fa/0/29	F3-IT -PC-29	F0/1	40	ACCESS
Fa/0/30	F3-IT -PC-30	F0/1	40	ACCESS
Fa/0/31	F3-IT -PC-31	F0/1	40	ACCESS
Fa/0/32	F3-IT -PC-32	F0/1	40	ACCESS
Fa/0/33	F3-IT -PC-33	F0/1	40	ACCESS
Fa/0/34	F3-IT -PC-34	F0/1	40	ACCESS
Fa/0/35	F3-IT -PC-35	F0/1	40	ACCESS
Fa/0/36	F3-IT -PC-36	F0/1	40	ACCESS
Fa/0/37	F3-IT -PC-37	F0/1	40	ACCESS
Fa/0/38	F3-IT -PC-38	F0/1	40	ACCESS
Fa/0/39	F3-IT -PC-39	F0/1	40	ACCESS
Fa/0/40	F3-IT -PC-40	F0/1	40	ACCESS
Fa/0/41	F3-IT-PC-42	F0/1	40	ACCESS
Fa/0/42	F3-IT-PC-43	F0/1	40	ACCESS
Fa/0/43	F3-IT-PC-44	F0/1	40	ACCESS
Fa/0/44	F3-IT-PC-45	F0/1	40	ACCESS
Fa/0/45	F3-ITPC-46	F0/1	40	ACCESS
Fa/0/46	F3-IT-PC-47	F0/1	40	ACCESS
Fa/0/47	F3-IT-PC-48	F0/1	40	ACCESS
Fa/0/48	F3-IT -PC-49	F0/1	40	ACCESS
GigabitEthernet 0/1	CO-SW-1	GigabitEthernet 0/11	-	TRUNK
GigabitEthernet 0/2	CO-SW-2	GigabitEthernet 0/11	-	TRUNK

Table 08 – Third Floor – SW_01

Third Floor- SW_02

INTERFACE	CONNECTS TO	CONNECTING PORTS	VLAN NO	PORT MODE
Fa/0/1	F3-IT-PC-50	F0/1	40	ACCESS
Fa/0/2	F3-IT-PC-51	F0/1	40	ACCESS
Fa/0/3	F3-IT-PC-52	F0/1	40	ACCESS
Fa/0/4	F3-IT-PC-53	F0/1	40	ACCESS
Fa/0/5	F3-IT-PC-54	F0/1	40	ACCESS
Fa/0/6	F3-IT-PC-55	F0/1	40	ACCESS
Fa/0/7	F3-IT-PC-56	F0/1	40	ACCESS
Fa/0/8	F3-IT-PC-57	F0/1	40	ACCESS
Fa/0/9	F3-IT-PC-58	F0/1	40	ACCESS
Fa/0/10	F3-IT-PC-59	F0/1	40	ACCESS
Fa/0/11	F3-IT-PC-60	F0/1	40	ACCESS
Fa/0/12	F3-IT-PC-61	F0/1	40	ACCESS
Fa/0/13	F3-IT-PC-62	F0/1	40	ACCESS
Fa/0/14	F3-IT-PC-63	F0/1	40	ACCESS
Fa/0/15	F3-IT-PC-64	F0/1	40	ACCESS
Fa/0/16	F3-IT-PC-65	F0/1	40	ACCESS
Fa/0/17	F3-IT-PC-66	F0/1	40	ACCESS
Fa/0/18	F3-IT-PC-67	F0/1	40	ACCESS
Fa/0/19	F3-IT-PC-68	F0/1	40	ACCESS
Fa/0/20	F3-IT-PC-69	F0/1	40	ACCESS
Fa/0/21	F3-IT-PC-70	F0/1	40	ACCESS
Fa/0/22	F3-IT-PC-71	F0/1	40	ACCESS
Fa/0/23	F3-IT-PC-72	F0/1	40	ACCESS
Fa/0/24	F3-IT-PC-73	F0/1	40	ACCESS
Fa/0/25	F3-IT-PC-74	F0/1	40	ACCESS
Fa/0/26	F3-IT-PC-75	F0/1	40	ACCESS
Fa/0/27	F3-IT-PC-76	F0/1	40	ACCESS
Fa/0/28	F3-IT-PC-77	F0/1	40	ACCESS

Fa/0/29	F3-IT-PC-78	F0/1	40	ACCESS
Fa/0/30	F3-IT-PC-79	F0/1	40	ACCESS
Fa/0/31	F2-AU-PC-1	F0/1	41	ACCESS
Fa/0/32	F2-AU-PC-2	F0/1	41	ACCESS
Fa/0/33	F3-AU -PC-3	F0/1	41	ACCESS
Fa/0/34	F3-AU -PC-4	F0/1	41	ACCESS
Fa/0/35	F3-AU -PC-5	F0/1	41	ACCESS
Fa/0/36	F3-AU -PC-6	F0/1	41	ACCESS
Fa/0/37	F3-AU -PC-7	F0/1	41	ACCESS
Fa/0/38	F3-AU -PC-8	F0/1	41	ACCESS
Fa/0/39	F3-AU-PC-9	F0/1	41	ACCESS
Fa/0/40	F3-AU-PC-10	F0/1	41	ACCESS
Fa/0/41	F3-AU-PC-11	F0/1	41	ACCESS
Fa/0/42	F3-AU-PC-12	F0/1	41	ACCESS
Fa/0/43	F3-AU-PC-13	F0/1	41	ACCESS
Fa/0/44	F3-AU-PC-14	F0/1	41	ACCESS
Fa/0/45	F3-AU-PC-15	F0/1	41	ACCESS
Fa/0/46	F3-AU-PC-16	F0/1	41	ACCESS
Fa/0/47	F3-AU-PC-17	F0/1	41	ACCESS
Fa/0/48	F3-AU -PC-18	F0/1	41	ACCESS
GigabitEthernet 0/1	CO-SW-1	GigabitEthernet 0/12	-	TRUNK
GigabitEthernet 0/2	CO-SW-2	GigabitEthernet 0/12	-	TRUNK

Table 09 – Third Floor – SW_02

Third Floor – SW_03

INTERFACE	CONNECTS TO	CONNECTING PORTS	VLAN NO	PORT MODE
Fa/0/1	F3-AU-PC-19	F0/1	41	ACCESS
Fa/0/2	F3-AU-PC-20	F0/1	41	ACCESS
Fa/0/3	F3-AD-PC-1	F0/1	42	ACCESS
Fa/0/4	F3-AD-PC-2	F0/1	42	ACCESS
Fa/0/5	F3-AD-PC-3	F0/1	42	ACCESS
Fa/0/6	F3-AD-PC-4	F0/1	42	ACCESS
Fa/0/7	F3-AD-PC-5	F0/1	42	ACCESS
Fa/0/8	F3-AD-PC-6	F0/1	42	ACCESS
Fa/0/9	F3-AD-PC-7	F0/1	42	ACCESS
Fa/0/10	F3-AD-PC-8	F0/1	42	ACCESS
Fa/0/11	F3-AD-PC-8	F0/1	42	ACCESS
Fa/0/12	F3-AD-PC-9	F0/1	42	ACCESS
Fa/0/13	F3-AD-PC-10	F0/1	42	ACCESS
Fa/0/14	F3-AD-PC-11	F0/1	42	ACCESS
Fa/0/15	F3-ADPC-12	F0/1	42	ACCESS
Fa/0/16	F3-AD-PC-13	F0/1	42	ACCESS
Fa/0/17	F3-AD-PC-14	F0/1	42	ACCESS
Fa/0/18	F3-AD-PC-15	F0/1	42	ACCESS
Fa/0/19	F3-AD-PC-16	F0/1	42	ACCESS
Fa/0/20	F3-AD-PC-17	F0/1	42	ACCESS
Fa/0/21	F3-AD-PC-18	F0/1	42	ACCESS
Fa/0/22	F3-AD-PC-13	F0/1	42	ACCESS
Fa/0/23	F3-AD-PC-14	F0/1	42	ACCESS
Fa/0/24	F3-AD-PC-15	F0/1	42	ACCESS
Fa/0/25	F3-AD-PC-16	F0/1	42	ACCESS
Fa/0/26	F3-AD-PC-17	F0/1	42	ACCESS
Fa/0/27	F3-AD-PC-19	F0/1	42	ACCESS

Fa/0/28	F3-AD-PC-20	F0/1	42	ACCESS
Fa/0/29	F3-AD-PC-21	F0/1	42	ACCESS
Fa/0/30	F3-AD-PC-22	F0/1	42	ACCESS
Fa/0/31	F3-AD-PC-23	F0/1	42	ACCESS
Fa/0/32	F3-AD-PC-24	F0/1	42	ACCESS
Fa/0/33	F3-AD-PC-25	F0/1	42	ACCESS
Fa/0/34	F3-AD-PC-26	F0/1	42	ACCESS
Fa/0/35	F3-AD-PC- 27	F0/1	42	ACCESS
Fa/0/36	F3-AD-PC-28	F0/1	42	ACCESS
Fa/0/37	F3-AD-PC- 29	F0/1	42	ACCESS
Fa/0/38	F3-AD-PC-30	F0/1	42	ACCESS
Fa/0/39	F3-DB-PC-1	F0/1	43	ACCESS
Fa/0/40	F3-DB-PC- 2	F0/1	43	ACCESS
Fa/0/41	F3-DB-PC-3	F0/1	43	ACCESS
Fa/0/42	F3-DB-PC-4	F0/1	43	ACCESS
Fa/0/43	F3-DB-PC- 5	F0/1	43	ACCESS
Fa/0/44	F3-DB-PC-6	F0/1	43	ACCESS
Fa/0/45	F3-DB-PC-7	F0/1	43	ACCESS
Fa/0/46	F3-DT-PC-1	F0/1	44	ACCESS
GigabitEthernet 0/1	CO-SW-1	GigabitEthernet 0/15	-	TRUNK
GigabitEthernet 0/2	CO-SW-2	GigabitEthernet 0/15	-	TRUNK

Table 10 – Third Floor – SW_03

3.1.4.2 Core Layer Switches

1. **Core Layer SW_01**

Core Layer SW_01

INTERFACE	CONNECTS TO	CONNECTING PORTS	PORT MODE
GigabitEthernet 1/0/1	GF-SW-1	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/2	1F-SW-1	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/3	1F-SW-2	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/4	2F-SW-1	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/5	2F-SW-2	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/6	2F-SW-3	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/7	3F-SW-1	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/8	3F-SW-2	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/9	3F-SW-3	GigabitEthernet 0/1	Trunk

Table 13 – Core Layer SW_01

Core Layer SW_02

INTERFACE	CONNECTS TO	CONNECTING PORTS	PORT MODE
GigabitEthernet 1/0/1	GF-SW-1	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/2	1F-SW-1	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/3	1F-SW-2	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/4	2F-SW-1	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/5	2F-SW-2	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/6	2F-SW-3	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/7	3F-SW-1	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/8	3F-SW-2	GigabitEthernet 0/1	Trunk
GigabitEthernet 1/0/9	3F-SW-3	GigabitEthernet 0/1	Trunk

Table 14 – Core Layer SW_02

3.2 PHYSICAL TOPOLOGY DESIGN

3.2.1 GENERAL FLOOR PLANS

- The company is in a building with four floors, each floor contains several departments. The height of the floor is 4 meters.
- Every floor is interconnected with a wiring duct which is used to send all the cables from all floors. Company is spread across a land with 900m2 enhancing the capabilities to handle multiple departments in one building.
- An appropriate method for cabling and wireless connectivity is provided to facilitate the users in the company.
- Each department consists of several PCs and all these PCs will be under centralized management. Cables for these PCs will be carried out in cable trunks, and these are attached to the walls to create a well-organized network environment.
- Based on the number of devices in a floor, the number of switches and patch panels in a cabinet vary.
- Standard for caballing in the bellow diagrams: **Red color - Ethernet cabling.**
- In this network, data is transferred through Ethernet wires. Having ethernet cables for the network is significantly more advantageous because it is simple to establish and costs nothing.
- Cables need to be carried through rooms and floors via conduits. Because it's simple to maintain and will give the environment the appearance of being well-managed, these are typically mounted to the ceilings.
- These conduits assist access points in connecting to the appropriate cables.
- To store the cables that connect to computers and printers inside of rooms, plastic cable trunks are used. Extendable plastic tubes are utilized to store the wires that connect to cameras.

1. GROUND FLOOR

This floor consists of the following equipment's;

- Desktop Computers – **37**
- Personal Computers – **3**
- Access Point – **8**
- CCTV – **5**
- Printers – **2**
- Cabinet – **1**

The floor consists of the following units;

- Reception
- Security Room
- Inquiry
- Legal
- Interview Room
- Lobby

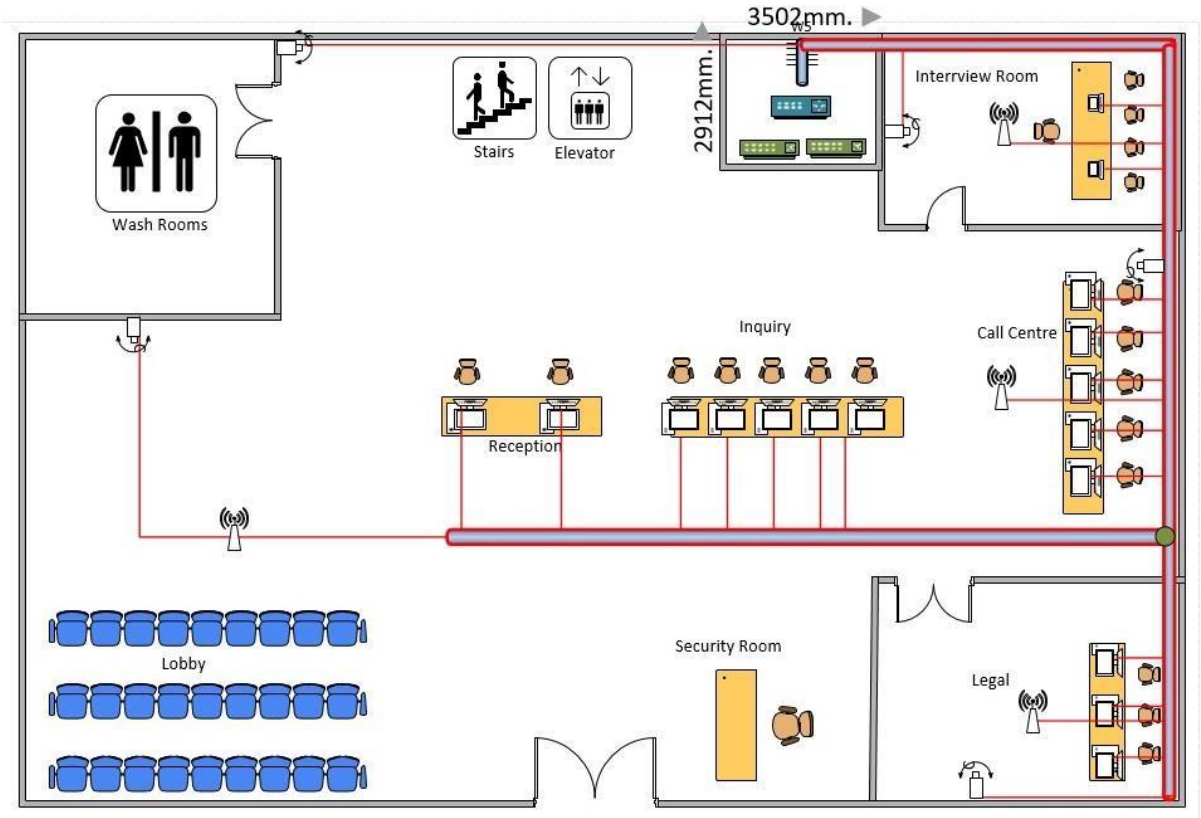


Figure 03 – Floor Plan of The Ground Floor

2. FIRST FLOOR

This floor consists of the following equipment's;

- Desktop Computers – **113**
- Personal Computers – **10**
- Access Point – **6**
- CCTV – **6**
- Printers – **4**
- Cabinet – **1**

The floor consists of the following units;

- Conference Room
- Finance Department
- Marketing Department
- HR Department
- Accounts Section

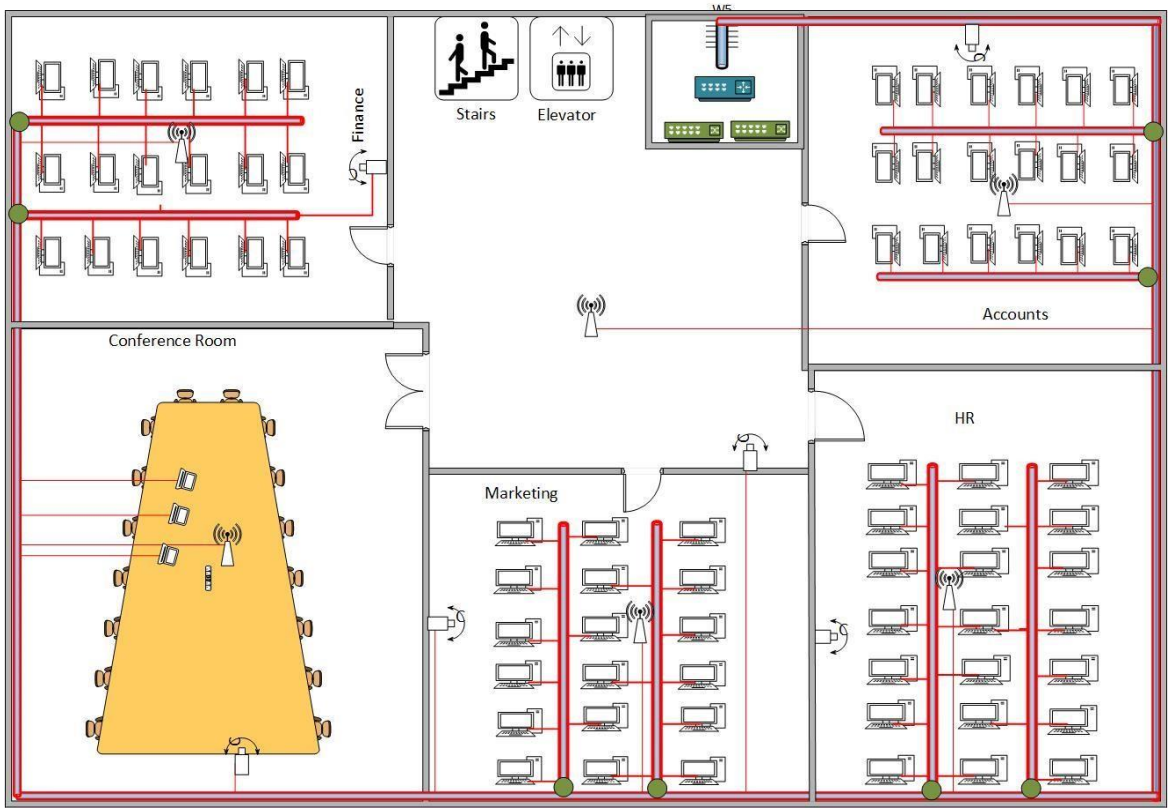


Figure 04 – Floor Plan of The First Floor

3. SECOND FLOOR

This floor consists of the following equipment's;

- Desktop Computers – **54**
- Personal Computers – **1**
- Access Point – **5**
- CCTV – **4**
- Printers – **4**
- Cabinet – **1**

The floor consists of the following units;

- Training Room
- Medical Centre
- Dining Area

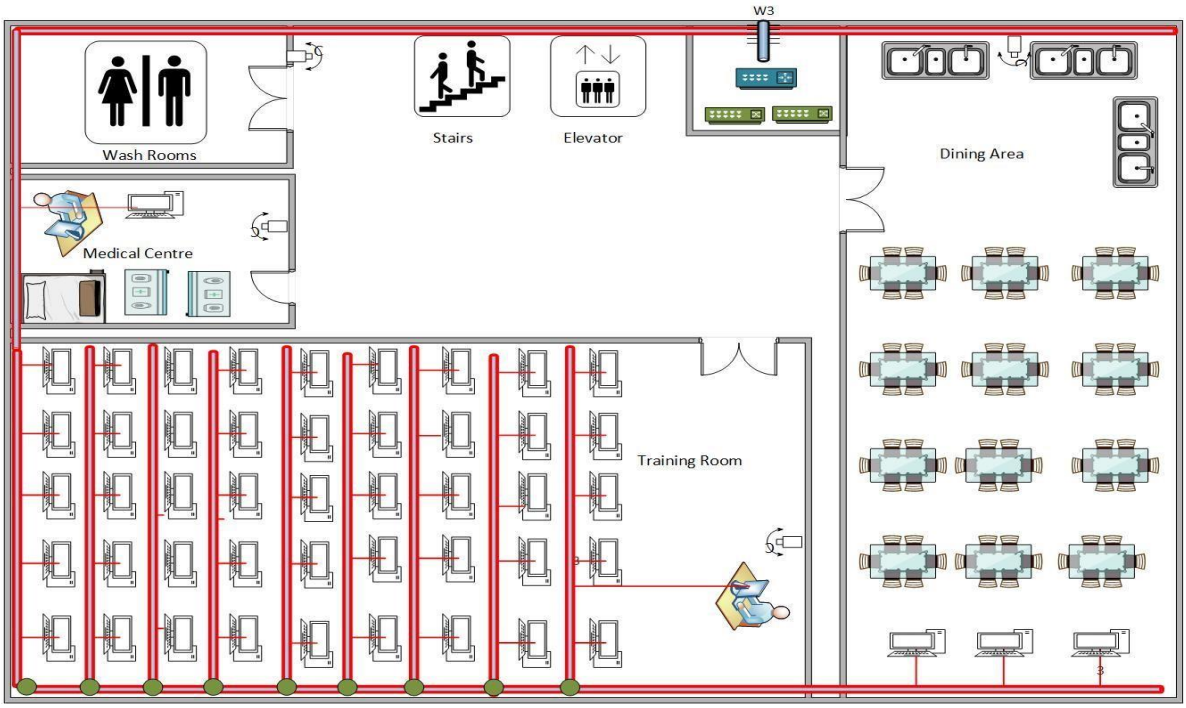


Figure 05 – Floor plan of The Second Floor

4.THIRD FLOOR

This floor consists of the following equipment's;

- Desktop Computers – **119**
- Personal Computers – **4**
- Access Point – **7**
- CCTV – **7**
- Printers – **4**
- Cabinet – **1**

The floor consists of the following units;

- Director Rooms
- Administrative Board
- Audit
- IT Department
- Data Room
- Server Room

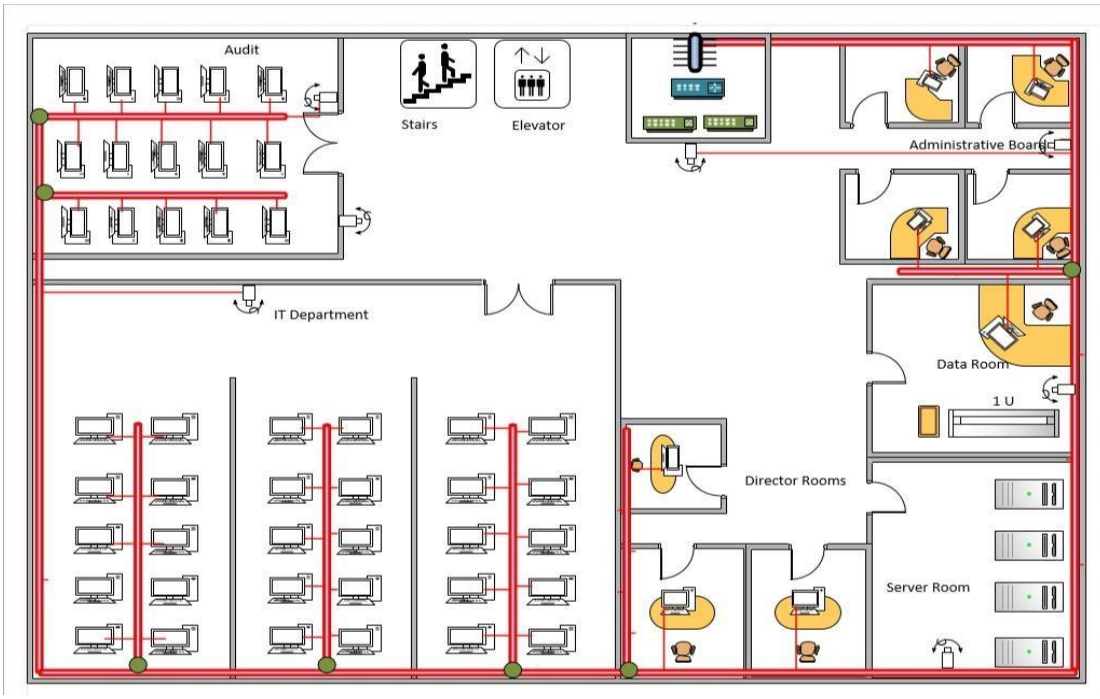


Figure 06 – Floor Plan of The Third Floor

3.2.2 PLAN FOR WIRELESS ACCESS POINTS

Floor	Quantity
Ground Floor	**8**
First Floor	**6**
Second Floor	**5**
Third Floor	**7**

Table 11 – Wireless Access Points Count for Each Floors

3.2.3 HEAT MAP DESIGNS FOR EACH FLOORS

1.GROUND FLOOR

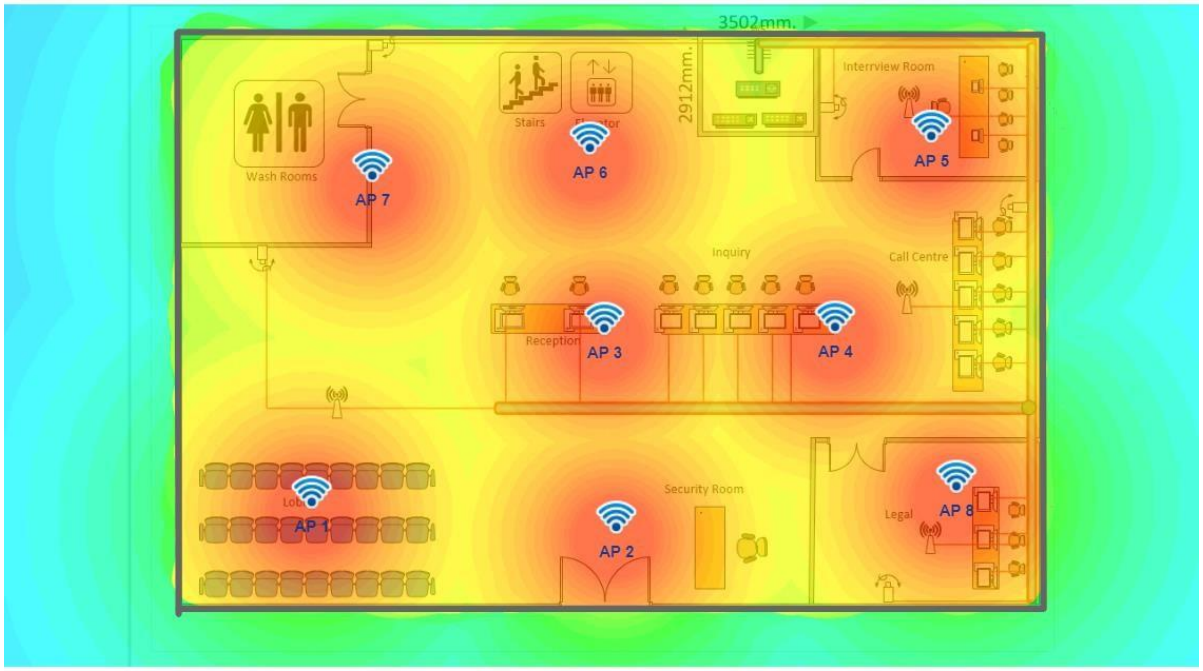


Figure 07 –Heat Map Design For The Ground Floor

2.FIRST FLOOR

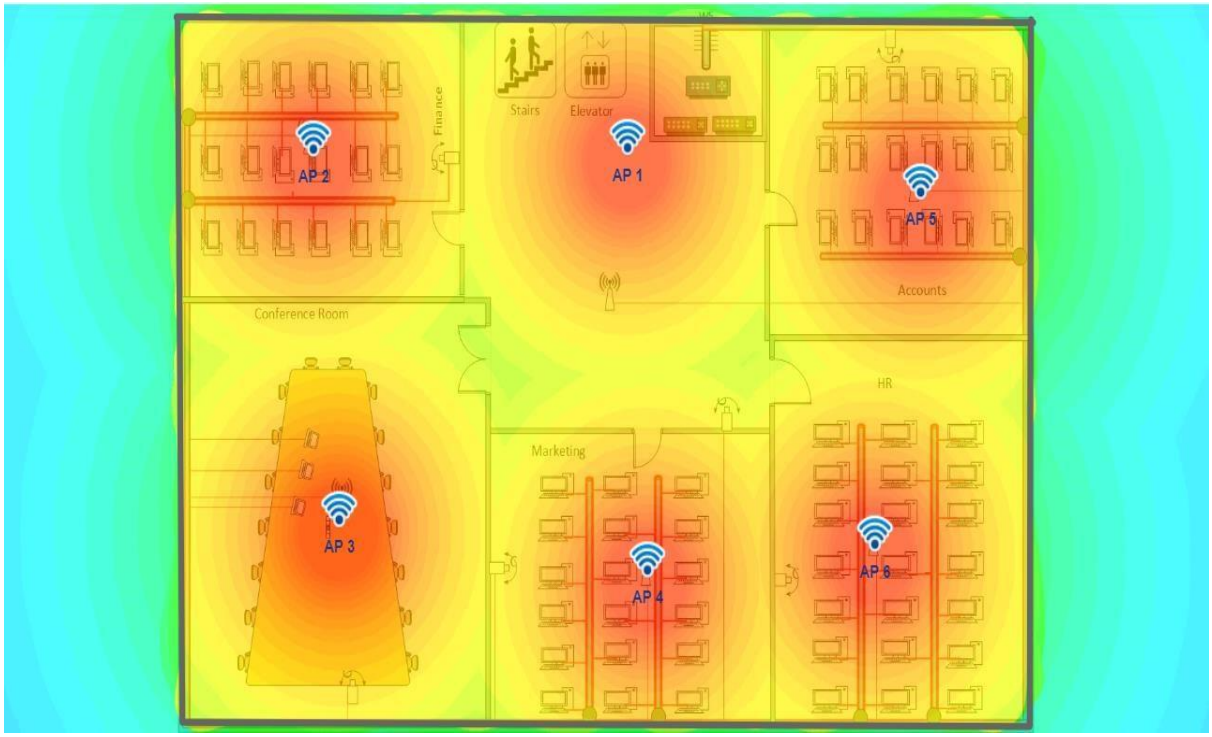


Figure 08 –Heat Map Designs For The First Floor

3. SECOND FLOOR



Figure 09 –Heat Map Designs For The Second Floor

4.THIRD FLOOR



Figure 10 –Heat Map Designs For The Third Floor

3.2.4 RACK DIAGRAM

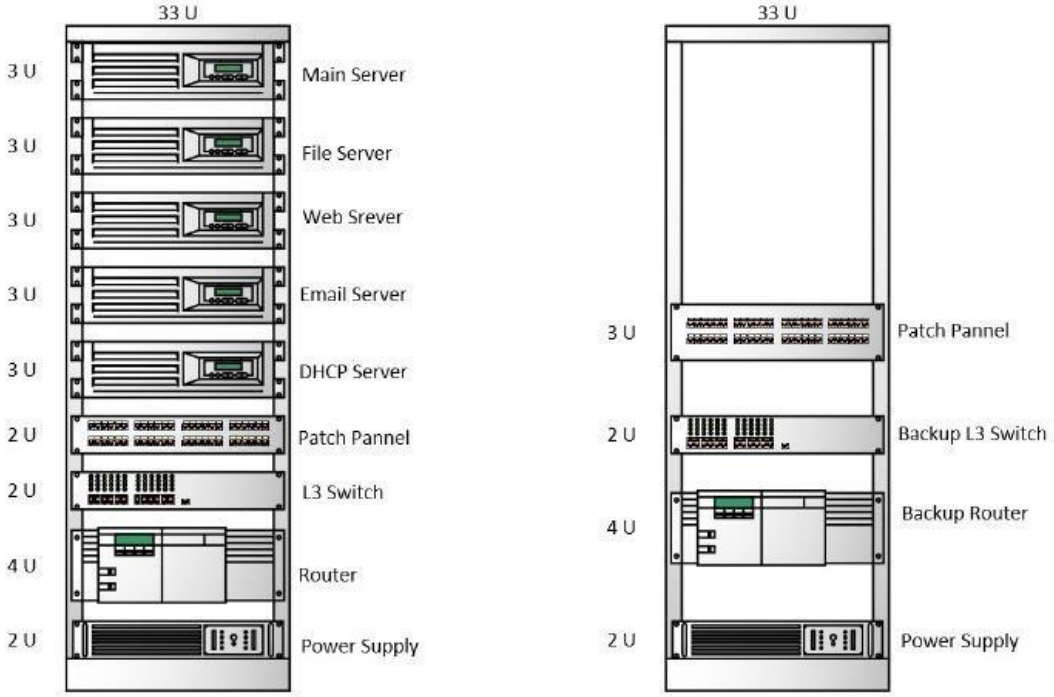


Figure 11 –Rack Diagram

3.2.5 CABINET DIAGRAMS

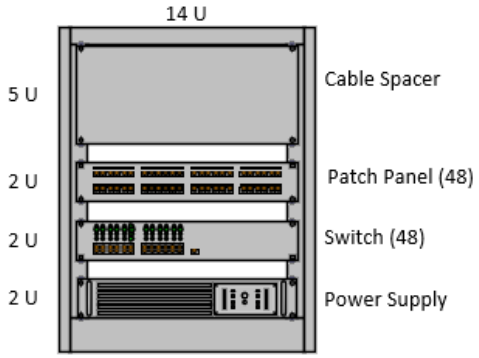


Figure 12 – Ground Floor Cabinet

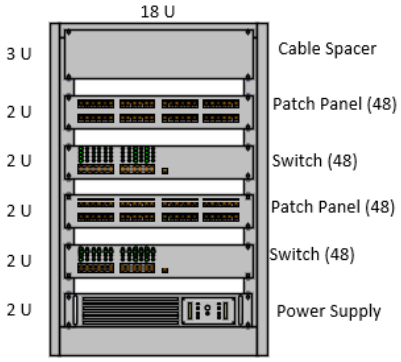


Figure 13 – First Floor Cabinet

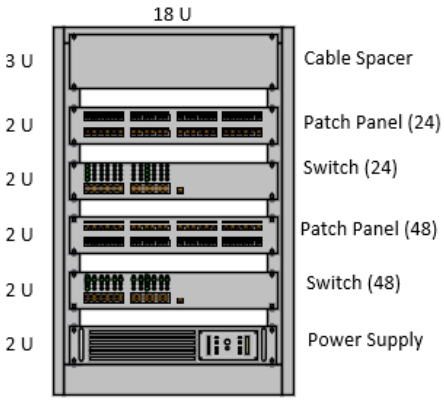


Figure 14 – First Floor Cabinet

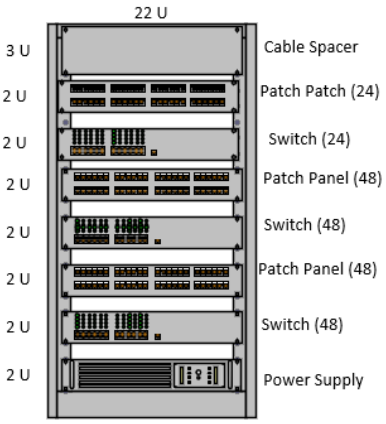


Figure 15 – First Floor Cabinet

3.3 EXPENDITURE

3.3.1 CALCULATE NETWORK BANDWIDTH

- The amount of bandwidth is determined by how each user uses the network. The expected bandwidth for the entire business will be provided by a primary leased line, with a secondary leased line being preserved for redundancy.

Floor	Department	Users	Bandwidth (MBps)	Total Bandwidth (MBps)	Backup Bandwidth (MBps)	Total Backup Bandwidth (MBps)
Ground Floor	Reception	2	0.2	0.4	0.2	0.4
	Inquiry	5	0.2	1	1	5
	Legal	5	0.2	1	1	5
	Interview Room	5	0.2	1	1	5
	Call center	20	2	40	1	20
First Floor	Conference Room	20	2	40	1	20
	Finance Department	35	2	70	0.2	7
	Marketing Department	28	2	56	0.2	11.2
	HR Department	35	2	70	0.2	7
	Accounts Section	20	1	20	0.1	2
Second Floor	Training Room	50	0.2	10	1	1
	Medical Centre	1	0.2	0.2	0.2	0.2
	Dining Area	3	0.2	0.6	0.2	0.6
	Audit	20	1	20	0.1	2
	Director Rooms	7	0.3	2.1	0.1	0.7

Third Floor	Administrative Board	30	1	30	1	30
	Data Room	1	0.2	0.2	0.2	0.2
	IT Department	78	4	312	0.3	23.4
	Wireless Users	500	2	1000	2	1000
	CCTV	22	0.5	11	0.2	4.4
	Switches	10	4	40	1	10

Table 12 – Network Bandwidth Calculation

CHAPTER 4 :

Implementation

4.1 NETWORK DEVICES CONFIGURATIONS

4.1.1 Layer 3 devices configuration

The Core layer of the network should provide users with access to essential network benefits. Both core layer switches have been configured to facilitate secure remote access through SSH and robust password protection. Rapid spanning tree protocol is in place to manage network devices and reduce downtime.VLANs are utilized in the internal network of the switches, controlled by VTP. At the access layer, switches are interlinked, while devices are connected at the core layer.

4.1.1.1 Core Switch 01 (L3_Main Switch)

- Change the hostname

```
Switch>
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname L3_Main
L3_Main(config)#
L3_Main(config)#
L3_Main(config)#
L3_Main(config)#
L3_Main(config)#
L3_Main(config)#
L3_Main(config)#
```

Figure 16 – Changing hostname – L3Main switch

- Configuring SSH

```
L3_Main(config)#
L3_Main(config)#
L3_Main(config)#username admin privilege 15 secret 123
L3_Main(config)#line vty 0 4
L3_Main(config-line)#transport input ssh
L3_Main(config-line)#ip domain-name nibm.lk
L3_Main(config)#crypto key generate rsa
The name for the keys will be: L3_Main.nibm.lk
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

L3_Main(config)#
*Mar 1 0:15:9.348: %SSH-5-ENABLED: SSH 1.99 has been enabled
L3_Main(config)#
```

Figure 17 – Configuring SSH – L3 Main Switch

- Creating VLANs

```
L3_Main(config)#
L3_Main(config)#vlan 10
L3_Main(config-vlan)#name Marketing_Dept
L3_Main(config-vlan)#vlan 20
L3_Main(config-vlan)#name Finance_Dept
L3_Main(config-vlan)#vlan 30
L3_Main(config-vlan)#name Machanical_Dept
L3_Main(config-vlan)#vlan 40
L3_Main(config-vlan)#name It_Dept
L3_Main(config-vlan)#vlan 50
L3_Main(config-vlan)#name Hr_Dept
L3_Main(config-vlan)#vlan 100
L3_Main(config-vlan)#name Server
L3_Main(config-vlan)#
```

Figure 18 – Creating Vlans – L3 Main Switch

- VTP Configuration

```
L3_Main(config)#
L3_Main(config)#
L3_Main(config)#
L3_Main(config)#vtp mode server
Device mode already VTP SERVER.
L3_Main(config)#vtp domain nibm.lk
Changing VTP domain name from NULL to nibm.lk
L3_Main(config)#
L3_Main(config)#
```

Figure 19 – Configuring VTP – L3 Main Switch

- VLAN Ip address assignment

```
L3_Main(config-vlan)#
L3_Main(config-vlan)#
L3_Main(config-vlan)#int vlan 10
L3_Main(config-if)#ip address 172.16.2.129 255.255.255.128
L3_Main(config-if)#int vlan 20
L3_Main(config-if)#ip address 172.16.1.129 255.255.255.128
L3_Main(config-if)#int vlan 30
L3_Main(config-if)#ip address 172.16.2.1 255.255.255.128
L3_Main(config-if)#int vlan 40
L3_Main(config-if)#ip address 172.16.0.1 255.255.255.0
L3_Main(config-if)#int vlan 50
L3_Main(config-if)#ip address 172.16.1.1 255.255.255.128
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINK-5-CHANGED: Interface Vlan30, changed state to up

%LINK-5-CHANGED: Interface Vlan40, changed state to up

%LINK-5-CHANGED: Interface Vlan50, changed state to up
```

Figure 20 – Configuring VLAN IPs – L3 Main Switch

- Trunking Ports

```
L3_Main(config)#
L3_Main(config)#interface range gigabitethernet 1/0/1-2
L3_Main(config-if-range)#switchport mode trunk


L3_Main(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan50, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up
```

Figure 21 – Trunking Ports – L3Main switch

- STP Configuration

```
L3_Main(config)#
L3_Main(config)#
L3_Main(config)#spanning-tree vlan 1 root Primary
L3_Main(config)#spanning-tree vlan 10 root Primary
L3_Main(config)#spanning-tree vlan 20 root Primary
L3_Main(config)#spanning-tree vlan 30 root Primary
L3_Main(config)#spanning-tree vlan 40 root Primary
L3_Main(config)#spanning-tree vlan 50 root Primary
L3_Main(config)#spanning-tree vlan 100 root Primary
L3_Main(config)#
L3_Main(config)#
```

Figure 22 – Configuring STP – L3Main switch

- Ether Channel Configuration

```
L3_Main(config)#
L3_Main(config)#interface range gigabitEthernet 1/0/23-24
L3_Main(config-if-range)#channel-group 1 mode desirable
L3_Main(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/24, changed state to up
```

Figure 23 – Configure EtherChannel – L3Main switch

- HSRP Configuration

```
L3_Main(config-if-range)#
L3_Main(config-if-range)#int vlan 10
L3_Main(config-if)#standby 1 ip 172.16.2.131
L3_Main(config-if)#standby 1 preempt
L3_Main(config-if)#standby 1 priority 150
L3_Main(config-if)#int vlan 20
L3_Main(config-if)#standby 1 ip 172.16.1.131
L3_Main(config-if)#standby 1 preempt
L3_Main(config-if)#standby 1 priority 150
L3_Main(config-if)#int vlan 30
L3_Main(config-if)#standby 1 ip 172.16.2.3
L3_Main(config-if)#standby 1 preempt
L3_Main(config-if)#standby 1 priority 150
L3_Main(config-if)#int vlan 40
L3_Main(config-if)#standby 1 ip 172.16.0.3
L3_Main(config-if)#standby 1 preempt
L3_Main(config-if)#standby 1 priority 150
L3_Main(config-if)#int vlan 50
L3_Main(config-if)#standby 1 ip 172.16.1.3
L3_Main(config-if)#standby 1 preempt
L3_Main(config-if)#standby 1 priority 150
L3_Main(config-if)#
```

Figure 24 – Configuring HSRP – L3Main switch

- IP Routing

```
L3_Main(config)#
L3_Main(config)#
L3_Main(config)#
L3_Main(config)#ip routing
L3_Main(config)#
L3_Main(config)#
```

Figure 25 – Configuring Ip routing – L3Main switch

4.1.1.2 Core Switch 01 (L3_Backup Switch)

Here, a secondary Layer 3 switch has been employed in place of the primary Layer 3 switch. This device has been used to route the backup path in case the primary path goes down

- Change the Hostname

```
Switch>
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname L3_Backup
L3_Backup(config)#
L3_Backup(config)#
```

Figure 26 – Changing Hostname – L3Backup Switch

- Configuring the initial setting with SSH

```
L3_Backup(config)#
L3_Backup(config)#username admin privilege 15 secret 123
L3_Backup(config)#line vty 0 4
L3_Backup(config-line)#login local
L3_Backup(config-line)#transport input ssh
L3_Backup(config-line)#ip domain-name nibm.lk
L3_Backup(config)#crypto key generate rsa
The name for the keys will be: L3_Backup.nibm.lk
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

L3_Backup(config)#
*Mar 1 0:8:31.320: %SSH-5-ENABLED: SSH 1.99 has been enabled
L3_Backup(config)#
```

Figure 27 – Configuring SSH – L3Backup Switch

- Assigning IP Address to VLAN

```
L3_Backup(config)#
L3_Backup(config)#int vlan 10
L3_Backup(config-if)#ip address 172.16.2.129 255.255.255.128
L3_Backup(config-if)#int vlan 20
L3_Backup(config-if)#ip address 172.16.1.129 255.255.255.128
L3_Backup(config-if)#int vlan 30
L3_Backup(config-if)#ip address 172.16.2.1 255.255.255.128
L3_Backup(config-if)#int vlan 40
L3_Backup(config-if)#ip address 172.16.0.1 255.255.255.0
L3_Backup(config-if)#int vlan 50
L3_Backup(config-if)#ip address 172.16.1.1 255.255.255.128
L3_Backup(config-if)#
```

Figure 28 – Configuring VLAN Ips – L2Backup Switch

- VTP Configuration

```
L3_Backup(config)#
L3_Backup(config)#vtp mode server
Device mode already VTP SERVER.
L3_Backup(config)#vtp domain nibm.lk
Changing VTP domain name from NULL to nibm.lk
L3_Backup(config)#
L3_Backup(config)#
```

Figure 29 – Configuring VTP – L3Backup Switch

- Trunking Ports

```
L3_Backup(config)#
L3_Backup(config)#interface range gigabitethernet 1/0/2-3
L3_Backup(config-if-range)#switchport mode trunk


L3_Backup(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/3, changed state to up

L3_Backup(config-if-range)#
```

Figure 30 – Trunking Ports – L3Backup Switch

- STP Configuration

```
L3_Backup(config)#
L3_Backup(config)#spanning-tree vlan 1 root secondary
L3_Backup(config)#spanning-tree vlan 10 root secondary
L3_Backup(config)#spanning-tree vlan 20 root secondary
L3_Backup(config)#spanning-tree vlan 30 root secondary
L3_Backup(config)#spanning-tree vlan 40 root secondary
L3_Backup(config)#spanning-tree vlan 50 root secondary
L3_Backup(config)#spanning-tree vlan 100 root secondary
L3_Backup(config)#
L3_Backup(config)#
```

Figure 31 – Configuring STP – L3Backup Switch

- EtherChannel Configuration

```
L3_Backup(config)#
L3_Backup(config)#interface range gigabitEthernet 1/0/23-24
L3_Backup(config-if-range)#channel-group 1 mode desirable
L3_Backup(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/23, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/23, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/24, changed state to up

L3_Backup(config-if-range)#
L3_Backup(config-if-range)#
```

Figure 32 – EtherChannel Configuration – L3Backup Switch

- IP Routing

```
L3_Backup(config)#
L3_Backup(config)#
L3_Backup(config)#ip routing
L3_Backup(config)#
```

Figure 33 – Configuring IP Routing – L3 Backup Switch

- HSRP Configuration

```
L3_Main(config-if-range)#
L3_Main(config-if-range)#int vlan 10
L3_Main(config-if)#standby 1 ip 172.16.2.131
L3_Main(config-if)#standby 1 preempt
L3_Main(config-if)#standby 1 priority 150
L3_Main(config-if)#int vlan 20
L3_Main(config-if)#standby 1 ip 172.16.1.131
L3_Main(config-if)#standby 1 preempt
L3_Main(config-if)#standby 1 priority 150
L3_Main(config-if)#int vlan 30
L3_Main(config-if)#standby 1 ip 172.16.2.3
L3_Main(config-if)#standby 1 preempt
L3_Main(config-if)#standby 1 priority 150
L3_Main(config-if)#int vlan 40
L3_Main(config-if)#standby 1 ip 172.16.0.3
L3_Main(config-if)#standby 1 preempt
L3_Main(config-if)#standby 1 priority 150
L3_Main(config-if)#int vlan 50
L3_Main(config-if)#standby 1 ip 172.16.1.3
L3_Main(config-if)#standby 1 preempt
L3_Main(config-if)#standby 1 priority 150
L3_Main(config-if)#
```

Figure 34 – Configuring HSRP – L3Backup Switch

- Port Security

```
L3_Backup(config)#
L3_Backup(config)#interface gigabitEthernet 1/0/3
L3_Backup(config-if)#switchport port-security maximum 5
L3_Backup(config-if)#switchport port-security violation restrict
L3_Backup(config-if)#
L3_Backup(config-if)#
```

Figure 35 – Configuring Port Security – L3Backup Switch

- IP Helper configuration

```
L3_Backup(config)#
L3_Backup(config)#int vlan 10
L3_Backup(config-if)#ip helper-address 192.168.10.1
L3_Backup(config-if)#int vlan 20
L3_Backup(config-if)#ip helper-address 192.168.10.1
L3_Backup(config-if)#int vlan 30
L3_Backup(config-if)#ip helper-address 192.168.10.1
L3_Backup(config-if)#int vlan 40
L3_Backup(config-if)#ip helper-address 192.168.10.1
L3_Backup(config-if)#int vlan 50
L3_Backup(config-if)#ip helper-address 192.168.10.1
L3_Backup(config-if)#
```

Figure 36 – IP helper Configuration – L3Backup Switch

4.1.2 LAYER 2 DEVICES CONFIGURATIONS

Users of the network will have easy access to the Access layer. Every access layer switch is configured with SSH for remote access and strong passwords. Additionally, port security has improved network security. Network downtime can be decreased with Rapid Spanning Tree port-fast. VLANs have been used to regulate the performance of network devices.

4.1.2.1 Switch (SW_01 / SW_02)

4.1.2.1.1 L2 SW_01

- Change the Hostname

```
Switch>
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW_01
SW_01(config)#
SW_01(config)#
```

Figure 37 – Changing Hostname –L2 Switch 01

- SSH configurations

```
SW_01(config)#
SW_01(config)#username admin privilege 15 secret 123
SW_01(config)#line vty 0 4
SW_01(config-line)#login local
SW_01(config-line)#transport input ssh
SW_01(config-line)#ip domain-name nibm.lk
SW_01(config)#crypto key generate rsa
The name for the keys will be: SW_01.nibm.lk
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

SW_01(config)#
*Mar 1 0:25:0.991: %SSH-5-ENABLED: SSH 1.99 has been enabled
SW_01(config)#
```

Figure 38 – Configuring SSH – L2 Switch 01

- Trunking Ports

```
SW_01(config)#
SW_01(config)#interface range fastEthernet 0/1-2
SW_01(config-if-range)#switchport mode trunk
SW_01(config-if-range)#
SW_01(config-if-range)#
```

Figure 39 – Trunking ports – L2 Switch 01

- Assigning VLAN Access to interfaces

```
SW_01(config)#
SW_01(config)#interface fastEthernet 0/5
SW_01(config-if)#switchport mode access
SW_01(config-if)#switchport access vlan 20
SW_01(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

SW_01(config-if)#
SW_01(config-if)#ex
SW_01(config)#
SW_01(config)#
SW_01(config)#interface fastEthernet 0/6
SW_01(config-if)#switchport mode access
SW_01(config-if)#switchport access vlan 30
SW_01(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up

SW_01(config-if)#
SW_01(config-if)#ex
SW_01(config)#
SW_01(config)#interface fastEthernet 0/7
SW_01(config-if)#switchport mode access
SW_01(config-if)#switchport access vlan 40
SW_01(config-if)#
```

Figure 40 – Assigning VLANs – L2 Switch 01

- Port Security

```
SW_01(config)#
SW_01(config)#
SW_01(config)#interface fastEthernet 0/4
SW_01(config-if)#switchport port-security maximum 5
SW_01(config-if)#switchport port-security violation restrict
SW_01(config-if)#
SW_01(config-if)#
```

Figure 41 – Configuring Port Security – L2 Switch 01

- VTP Configuration

```
SW_01(config)#
SW_01(config)#vtp mode client
Setting device to VTP CLIENT mode.
SW_01(config)#vtp domain nibm.lk
Changing VTP domain name from NULL to nibm.lk
SW_01(config)#
SW_01(config)#
```

Figure 42 – Configuring VTP– L2 Switch 01

4.1.2.1.1 L2 SW_02

- Change the Hostname

```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#hos
Switch(config)#hostname SW_02
SW_02(config)#
SW_02(config)#
```

Figure 43 – Changing Hostname – L2 Switch 02

- SSH configurations

```
SW_02(config)#
SW_02(config)#username admin privilege 15 secret 123
SW_02(config)#line vty 0 4
SW_02(config-line)#login local
SW_02(config-line)#transport input ssh
SW_02(config-line)#ip domain-name nibm.lk
SW_02(config)#crypto key generate rsa
% You already have RSA keys defined named SW_02.nibm.lk .
% Do you really want to replace them? [yes/no]: yes
The name for the keys will be: SW_02.nibm.lk
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

SW_02(config)#
*Mar 1 0:40:22.72: %SSH-5-ENABLED: SSH 1.99 has been enabled
SW_02(config)#
```

Figure 44 – Configuring SSH – L2 Switch 02

- Trunking Ports

```
SW_02(config-if)#
SW_02(config-if)#
SW_02(config-if)#interface range fastEthernet 0/1-2
SW_02(config-if-range)#switchport mode trunk
SW_02(config-if-range)#
SW_02(config-if-range)#
```

Figure 45 – Trunking ports – L2 Switch 02

- Assigning VLAN Access to interfaces

```
SW_02(config-if)#interface fastEthernet 0/4
SW_02(config-if)#switchport mode access
SW_02(config-if)#switchport access vlan 10
SW_02(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

SW_02(config-if)#
SW_02(config-if)#interface fastEthernet 0/5
SW_02(config-if)#switchport mode access
SW_02(config-if)#switchport access vlan 20
SW_02(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

SW_02(config-if)#interface fastEthernet 0/6
SW_02(config-if)#switchport mode access
SW_02(config-if)#switchport access vlan 30
SW_02(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up

SW_02(config-if)#interface fastEthernet 0/7
SW_02(config-if)#switchport mode access
SW_02(config-if)#switchport access vlan 40
SW_02(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up
```

Figure 46 – Assigning VLANs – L2 Switch 02

- Port Security

```
SW_02(config)#
SW_02(config)#interface fastEthernet 0/4
SW_02(config-if)#switchport port-security maximum 5
SW_02(config-if)#switchport port-security violation restrict
SW_02(config-if)#
SW_02(config-if)#
```

Figure 47 – Configuring Port Security – L2 Switch 02

- VTP Configuration

```
SW_02(config)#
SW_02(config)#vtp mode client
Setting device to VTP CLIENT mode.
SW_02(config)#vtp domain nibm.lk
Changing VTP domain name from NULL to nibm.lk
SW_02(config)#
```

Figure 48 – Configuring VTP– L2 Switch 02

4.1.3 ROUTER CONFIGURATIONS

- Initial Security Configurations

```
Router01(config)#
Router01(config)#ip domain name nibm.lk
Router01(config)#crypto key generate rsa
The name for the keys will be: Router01.nibm.lk
Choose the size of the key modulus in the range of 360 to 2048 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Router01(config)#enable password cisco
*Mar 1 0:2:5.586: %SSH-5-ENABLED: SSH 1.99 has been enabled
Router01(config)#username admin password admin
Router01(config)#ip ssh version 2
Router01(config)#line vty 0 15
Router01(config-line)#transport input ssh
Router01(config-line)#login local
Router01(config-line)#exit
Router01(config)#
```

Figure 49 – Initial Security Configuration – Router

- Change the hostname

```
Router>
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router_01
Router_01(config)#
Router_01(config)#
```

Figure 50 – Changing Hostname – Router

- EIGRP Configuration

```
Router_01(config)#router eigrp 1
Router_01(config-router)#net
Router_01(config-router)#network 172.16.5.1 0.0.0.127
Router_01(config-router)#net
Router_01(config-router)#network 172.16.0.0 0.0.0.0
Router_01(config-router)#
Router_01(config-router)#
```

Figure 51 – Configuring EIGRP – Router

- NAT Configurations

```
Router_01(config)#acc
Router_01(config)#access-list 1 permit 172.16.10.0 0.0.0.255
Router_01(config)#access-list 1 permit 172.16.20.0 0.0.0.255
Router_01(config)#access-list 1 permit 172.16.30.0 0.0.0.255
Router_01(config)#access-list 1 permit 172.16.40.0 0.0.0.255
Router_01(config)#access-list 1 permit 172.16.50.0 0.0.0.255
Router_01(config)#access-list 1 permit 172.16.0.0 0.0.0.0
Router_01(config)#
```

Figure 52 – NAT Configuration – Router

- NAT Overload Configurations

```
Router_01(config)#
Router_01(config)#
Router_01(config)#ip nat inside source list 1 interface g0/1 overload
Router_01(config)#
Router_01(config)#
```

Figure 53 – PAT Configuration – Router

- ACL Configurations

```
Router_01(config)#acc
Router_01(config)#access-list 1 permit 172.16.10.0 0.0.0.255
Router_01(config)#access-list 1 permit 172.16.20.0 0.0.0.255
Router_01(config)#access-list 1 permit 172.16.30.0 0.0.0.255
Router_01(config)#access-list 1 permit 172.16.40.0 0.0.0.255
Router_01(config)#access-list 1 permit 172.16.50.0 0.0.0.255
Router_01(config)#access-list 1 permit 172.16.0.0 0.0.0.0
Router_01(config)#
```

Figure 54 – ACL Configuration – Router

- IP Address Assigning

```
Router01(config)#
Router01(config)#
Router01(config)#interface gigabitEthernet 0/1
Router01(config-if)#ip address 172.16.5.3 255.255.255.0
```

Figure 55 – IP Address Configuration - Router

4.2 SERVER CONFIGURATION

The network server has been equipped with Microsoft Windows Server 2016, which will serve as the foundation for deploying various domain administration applications. This platform has been chosen for its ability to manage and control multiple operational aspects seamlessly. By utilizing specific Windows Server 2016 roles tailored to our needs, we can enhance efficiency, security, and scalability throughout our network environment.

- Active Directory Domain Service (ADDS)
- DHCP
- DNS
- File Management Services

4.2.1 Initial Configurations

- New Virtual Machine Wizard

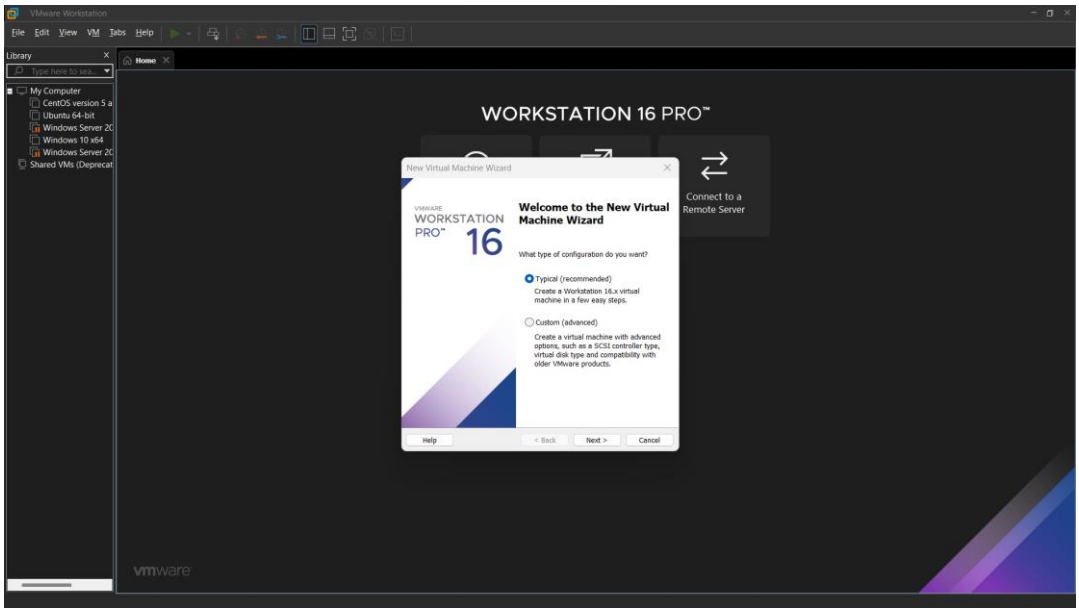


Figure 56 - New Virtual Machine Wizard- Windows Server 2016

- Guest Operating System Installation

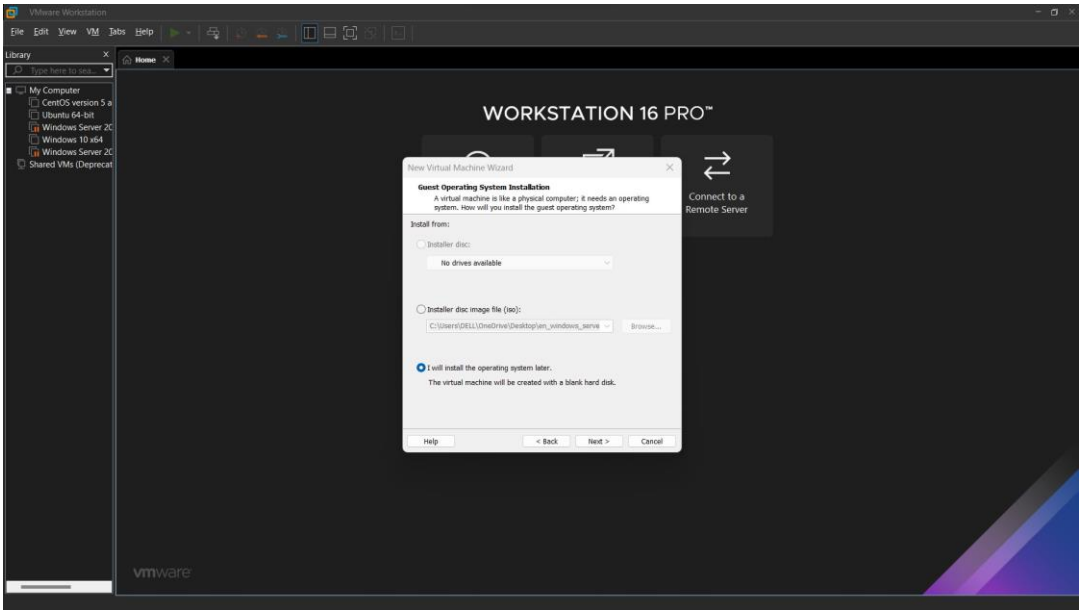


Figure 57 - Guest Operating System Installation - Windows Server 2016

- Selecting a Guest Operating System

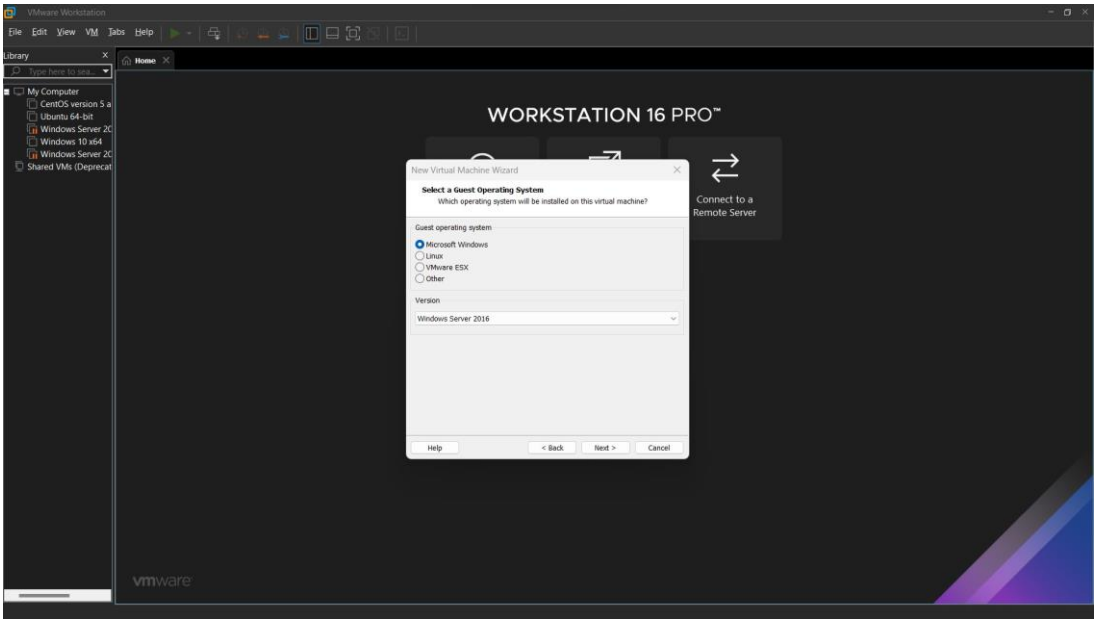


Figure 58 - Selecting a Guest Operating System- Windows Server 2016

- Name the Virtual Machine

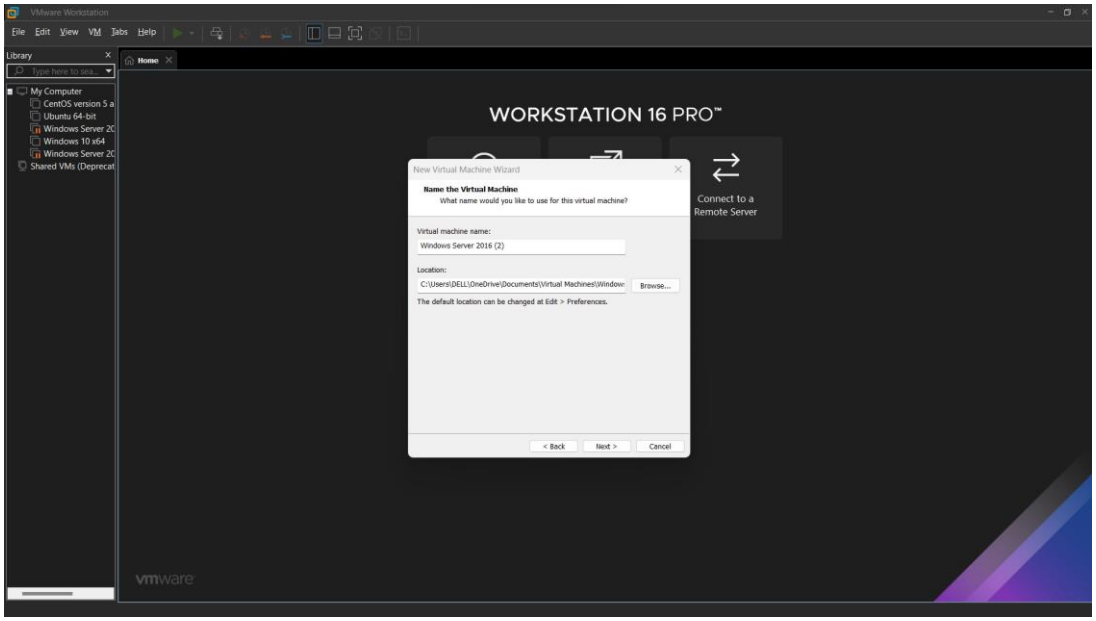


Figure 59 - Name the Virtual Machine - Windows Server 2016

- Specify Disk Capacity

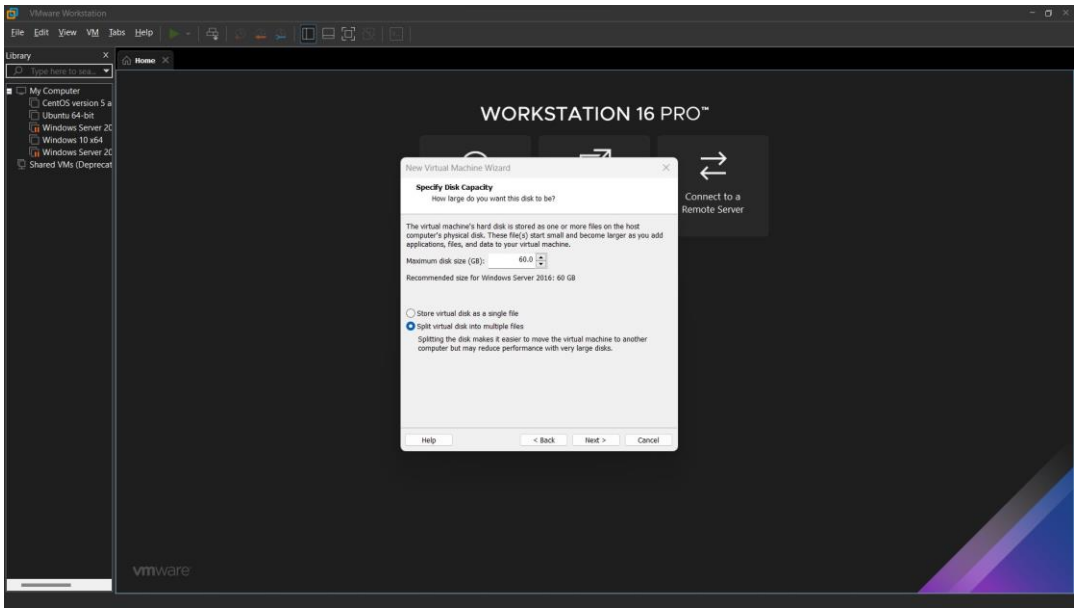


Figure 60 - Specify Disk Capacity - Windows Server 2016

- Ready to Create Virtual Machine

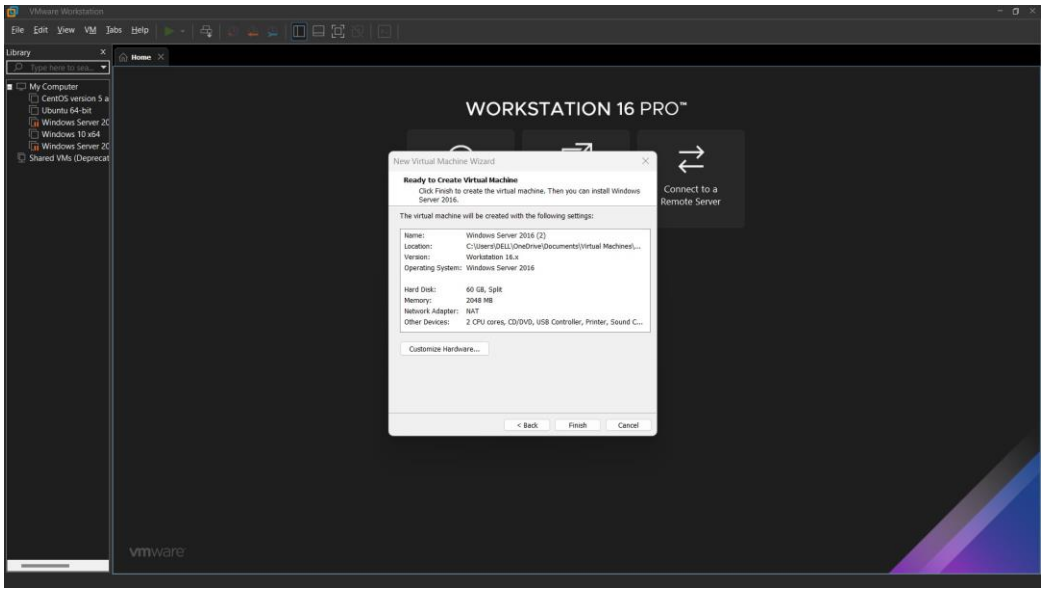


Figure 61 - Ready to Create Virtual Machine - Windows Server 2016

- Installing windows

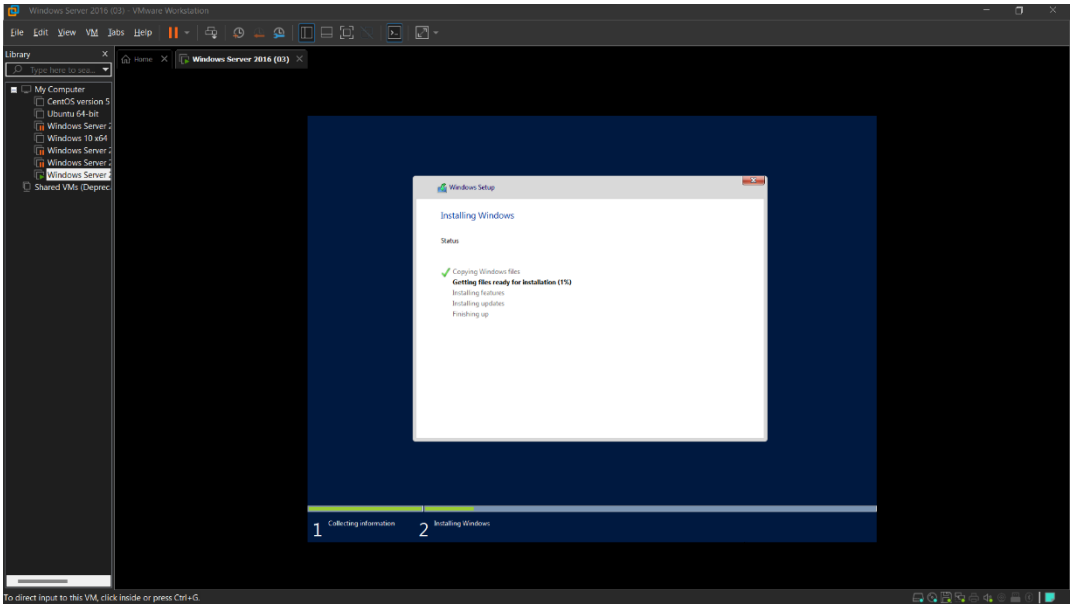


Figure 62 - Installing windows - Windows Server 2016

- Windows Setup

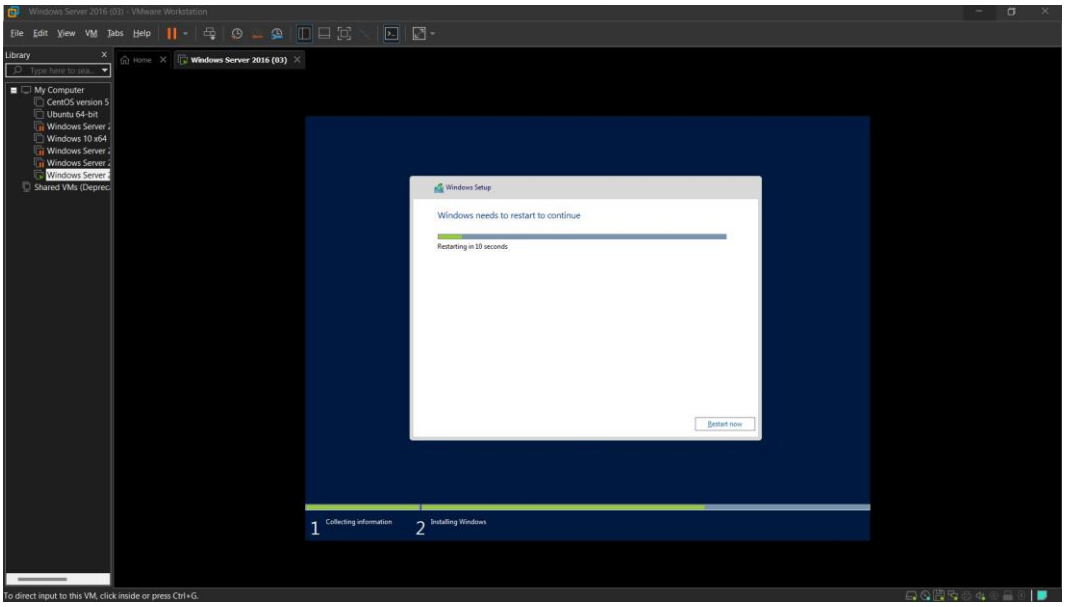


Figure 63 - Windows Setup - Windows Server 2016

- VMware Tools Setup

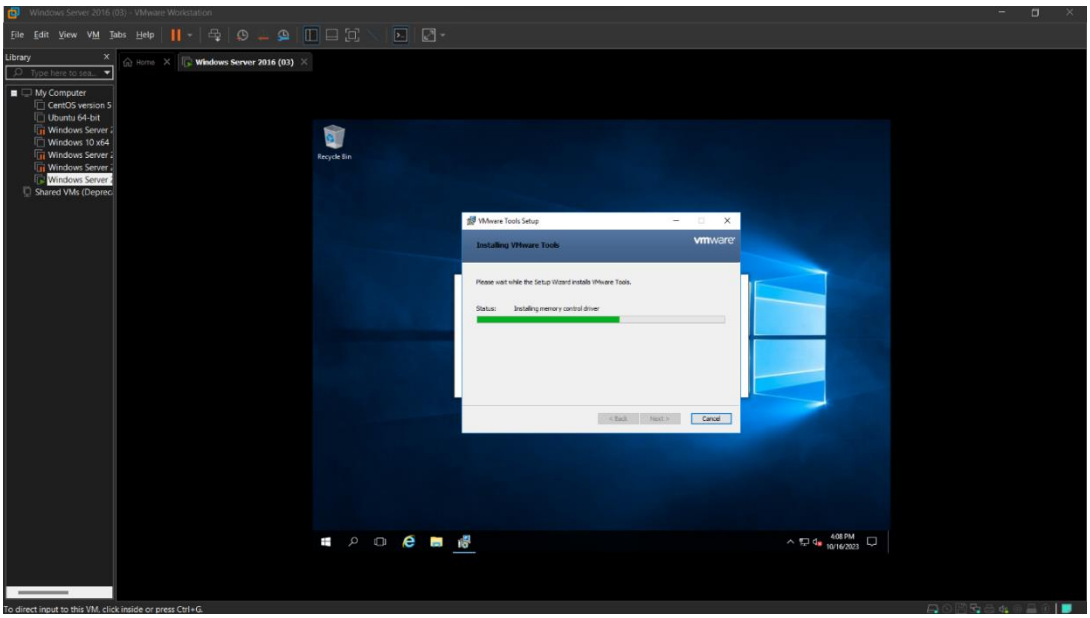


Figure 64 - VMware Tools Setup - Windows Server 2016

- Server Manager

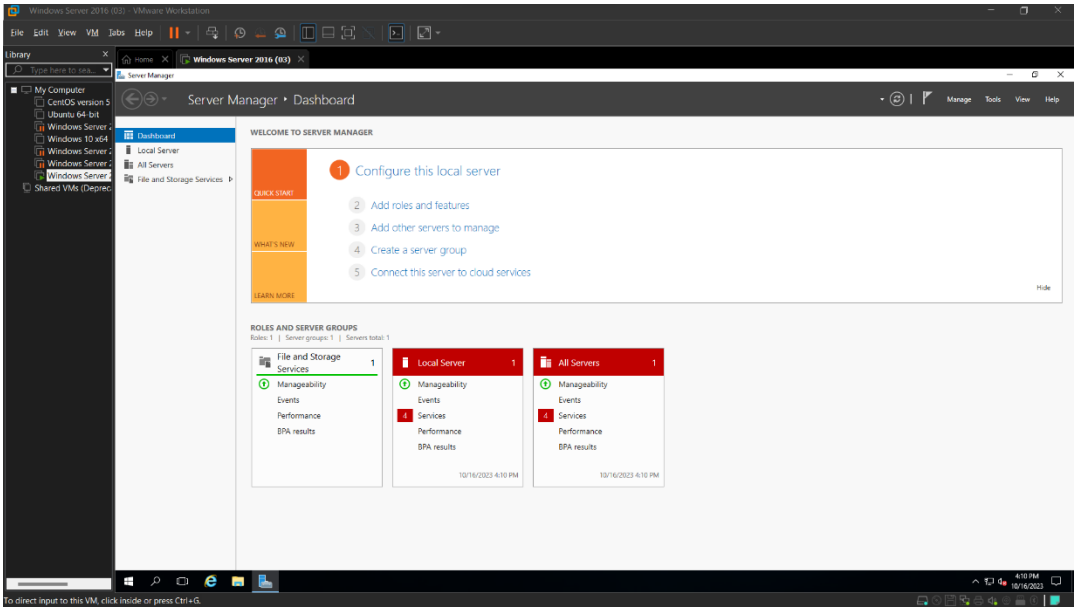


Figure 65 - Server Manager - Windows Server 2016

- Select DHCP, DNS, ADDS, Roles for installation and add features

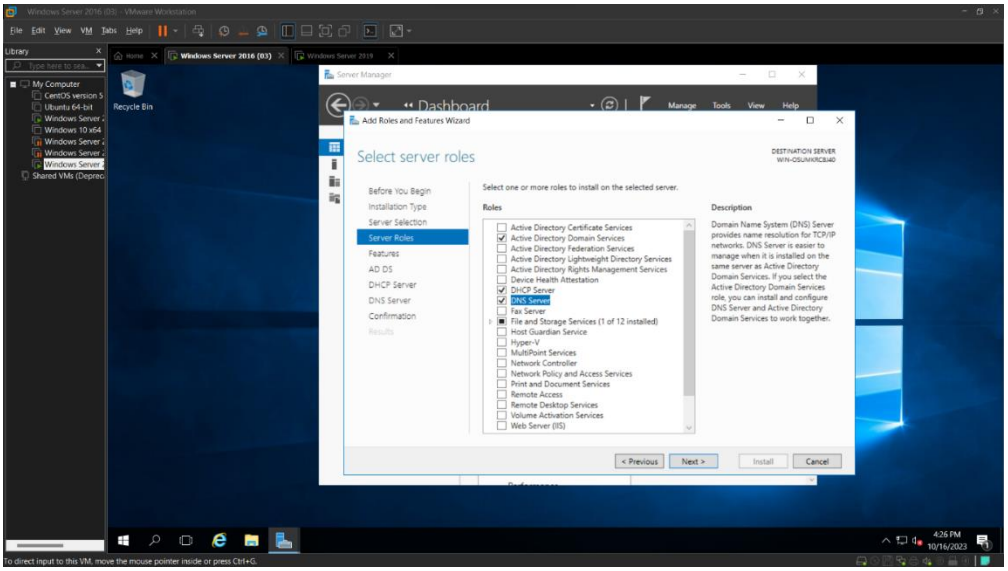


Figure 66 – Installation and add features for DHCP, DNS, ADDS Roles

- Confirm Installion Selection

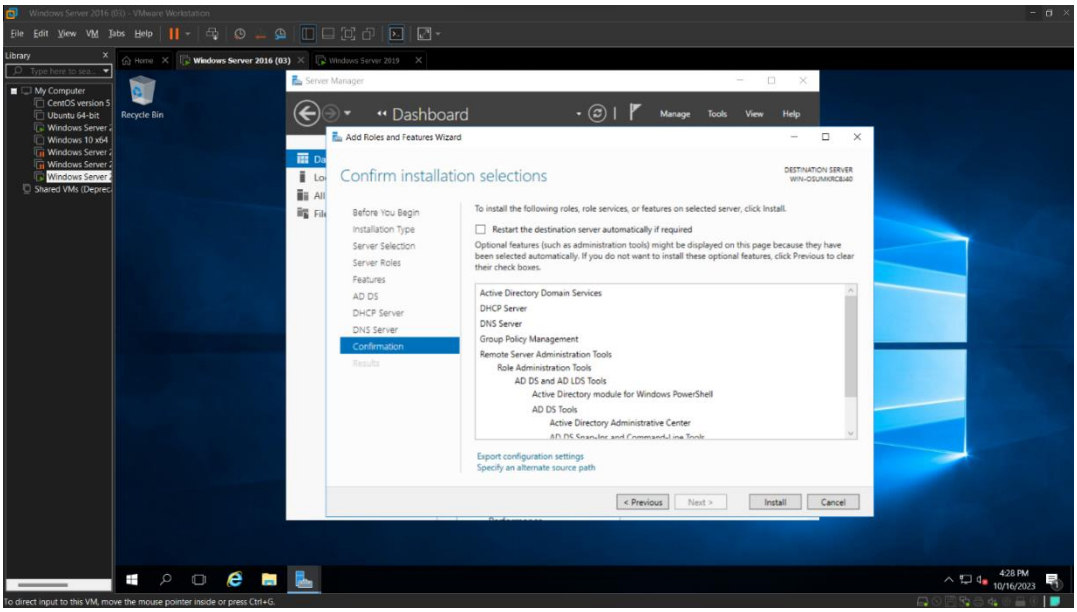


Figure 67 - Confirm Installion Selection - Windows Server 2016

- Installation of DHCP, DNS, ADDS Roles

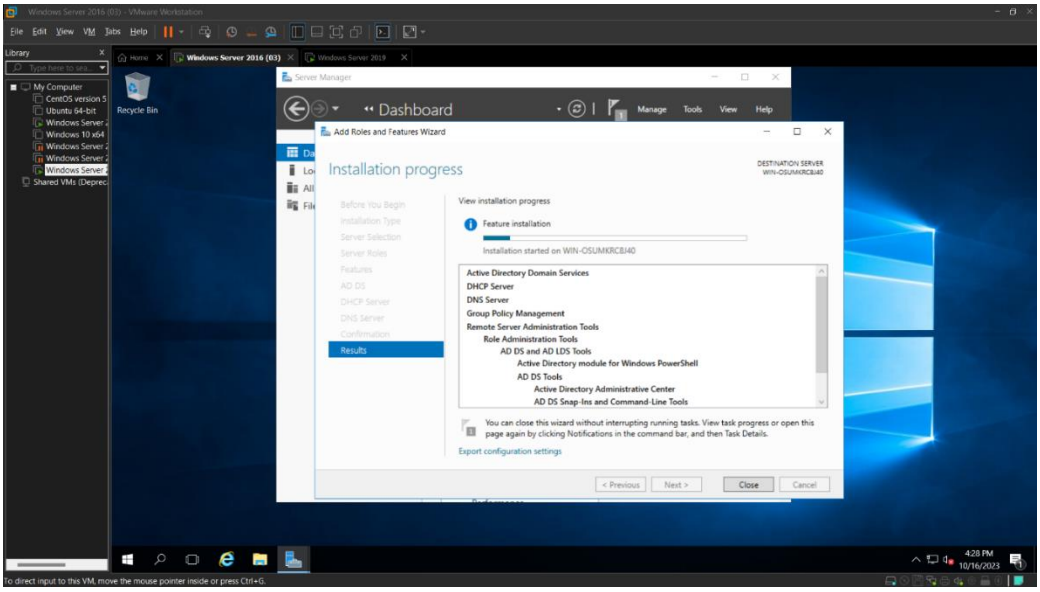


Figure 68 – Confirm Installation selection - Windows Server 2016

- Installation of DHCP, DNS, ADDS Roles

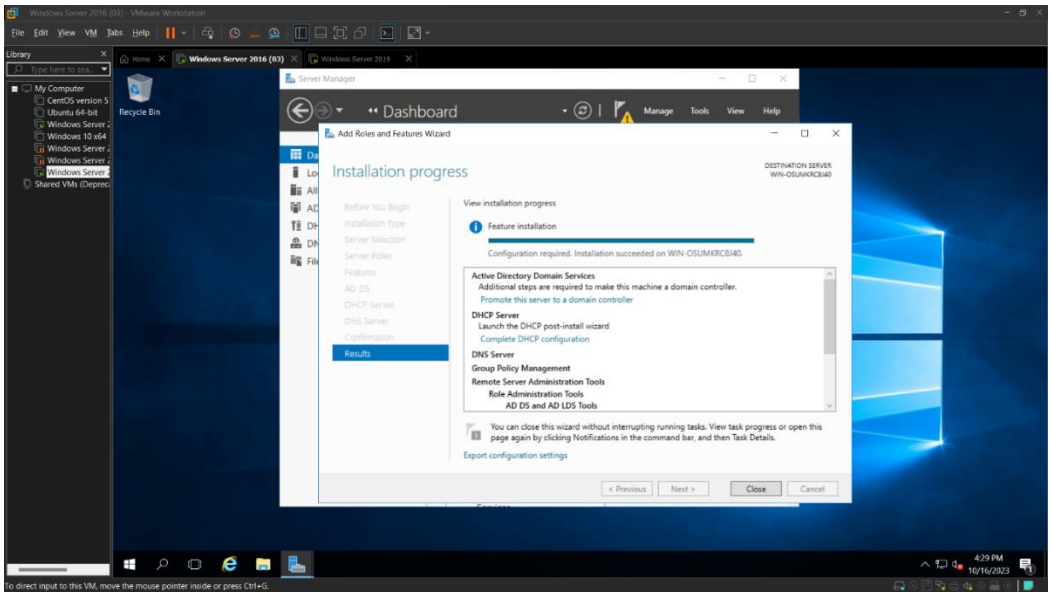


Figure 69–Installation of DHCP,DNS,ADDS Roles-Windows Server 2016

4.2.2 Add Server Role Configurations

- ADDS Deployment Configuration

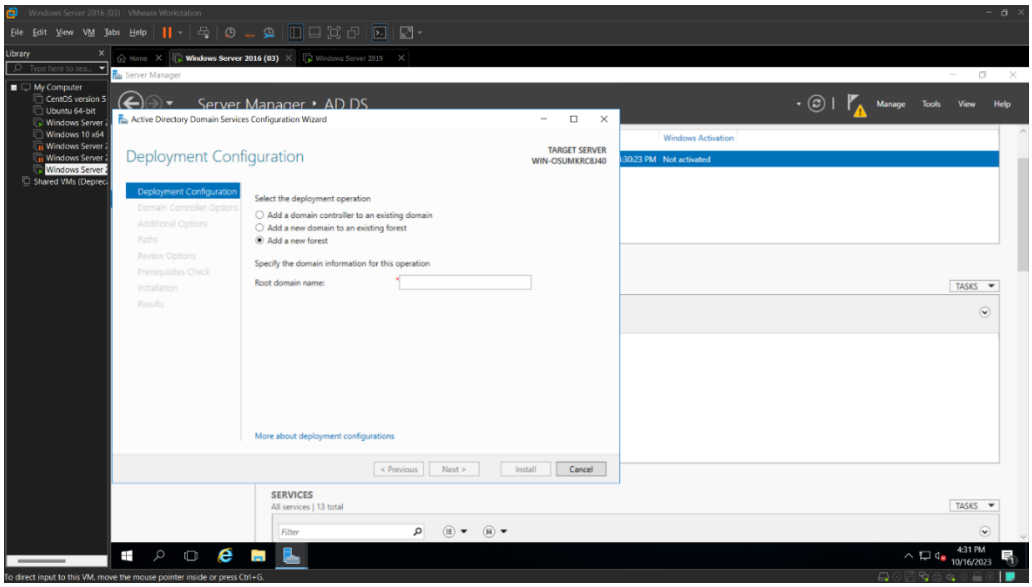


Figure 70 – ADDS Deployment Configuration (i) - Windows Server 2016

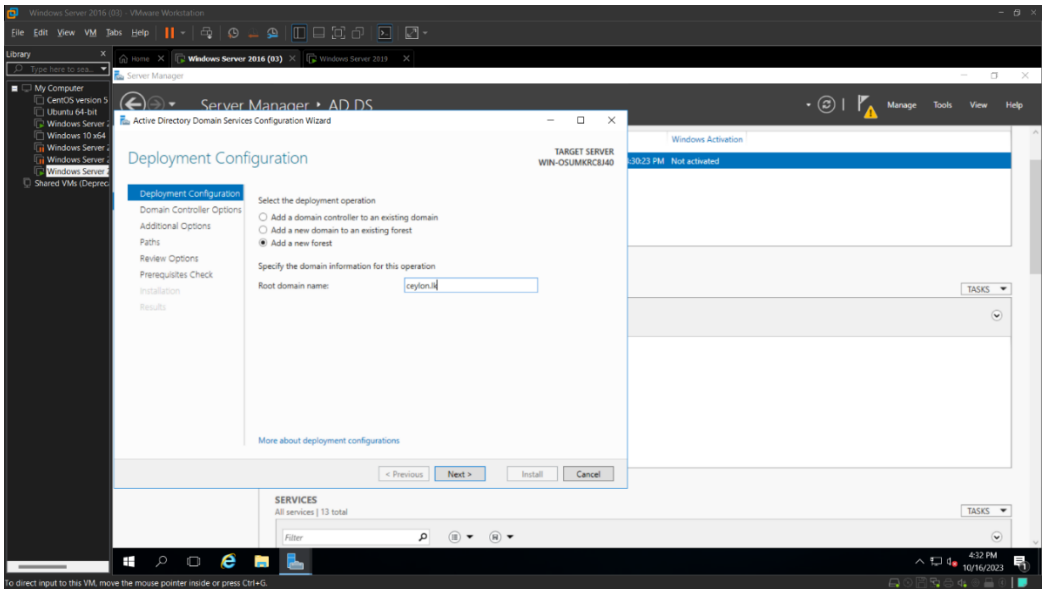


Figure 71 – ADDS Deployment Configuration (ii) - Windows Server 2016

- Domain Controller Option

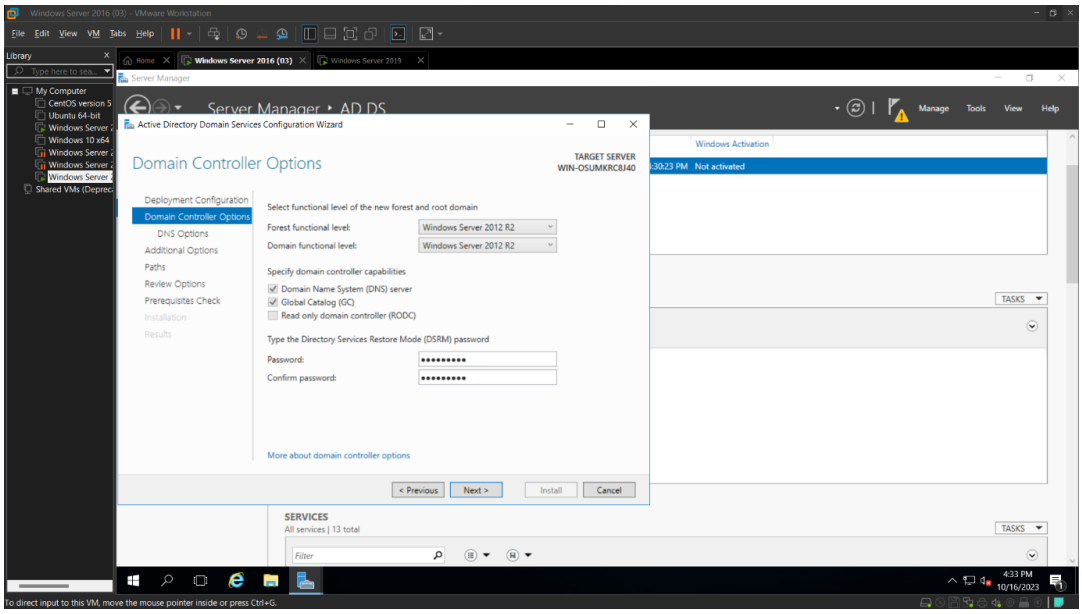


Figure 72 – Domain Controller Option - Windows Server 2016

- Prerequisites Check for ADDS

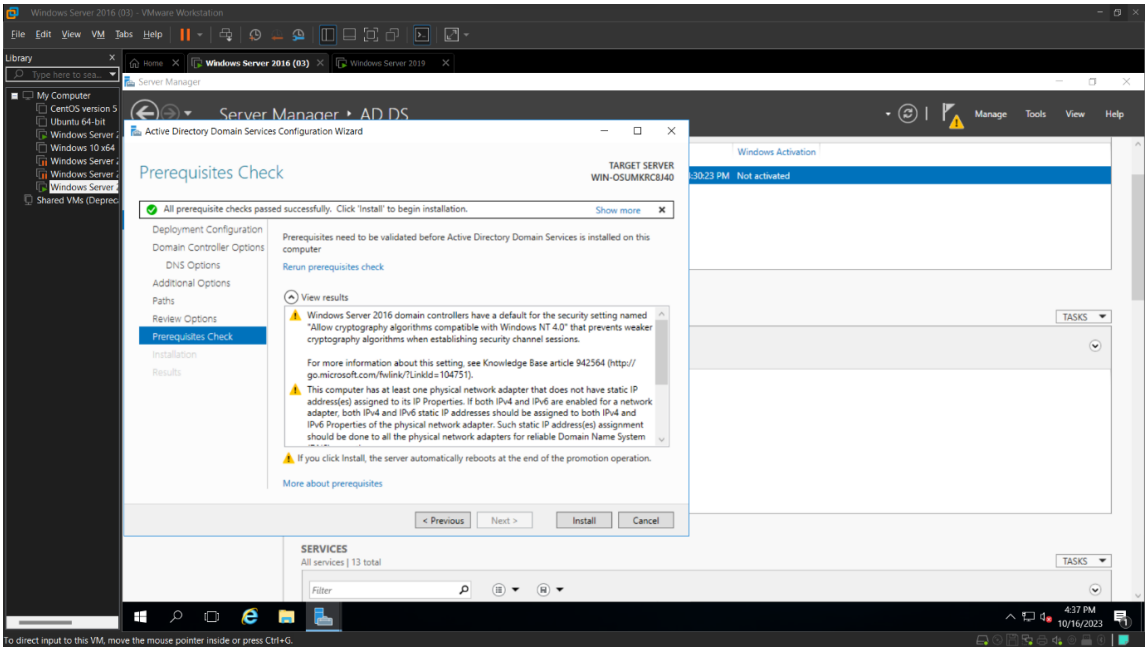


Figure 73 – Prerequisites Check for ADDS - Windows Server 2016

- Installation

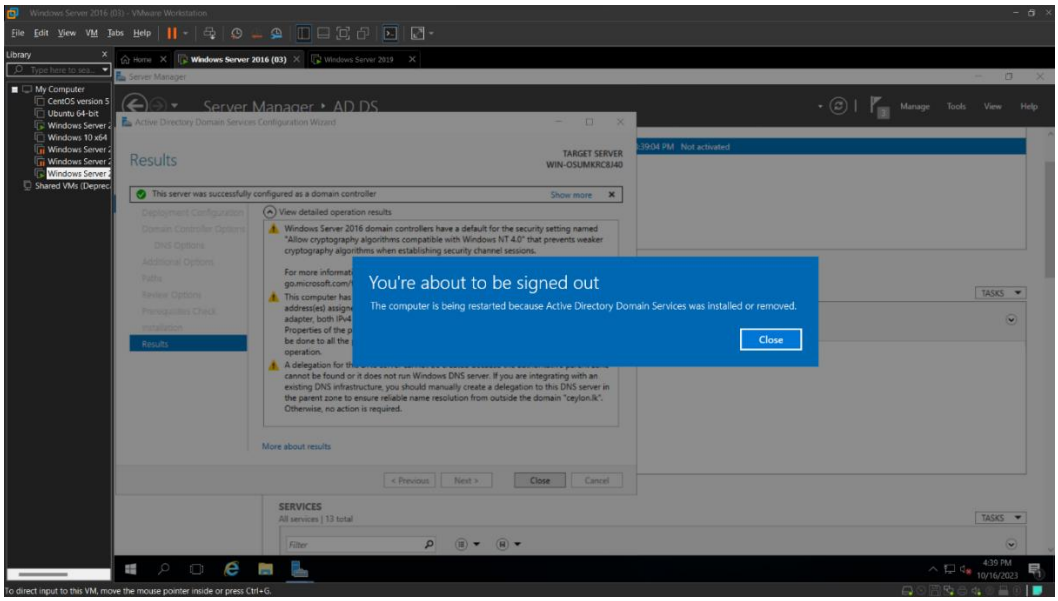


Figure 74 – Installation - Windows Server 2016

- Verification

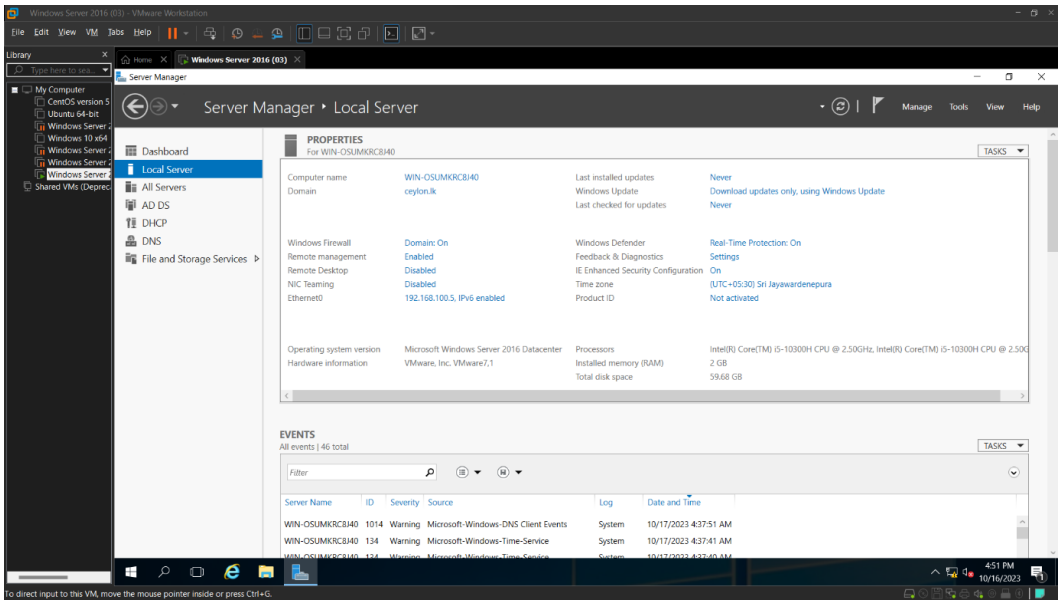


Figure 75 – Verification - Windows Server 2016

4.2.3 DHCP SERVER CONFIGURATION

- Post-Deployment configuration for DHCP

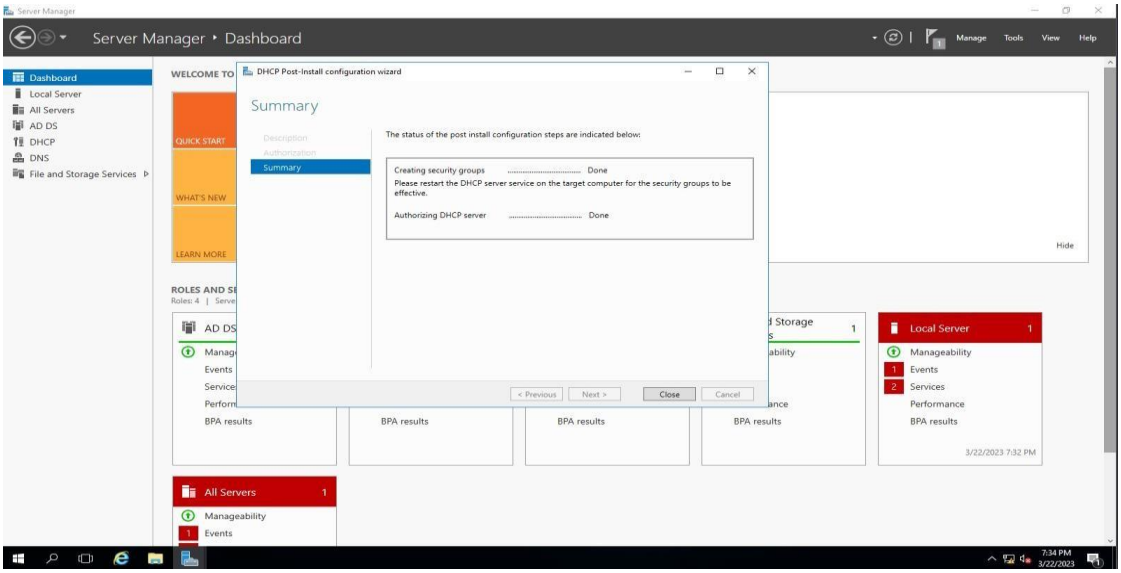


Figure 76 - Post-Deployment configuration for DHCP - Windows Server 2016

- Creating DHCP Scops

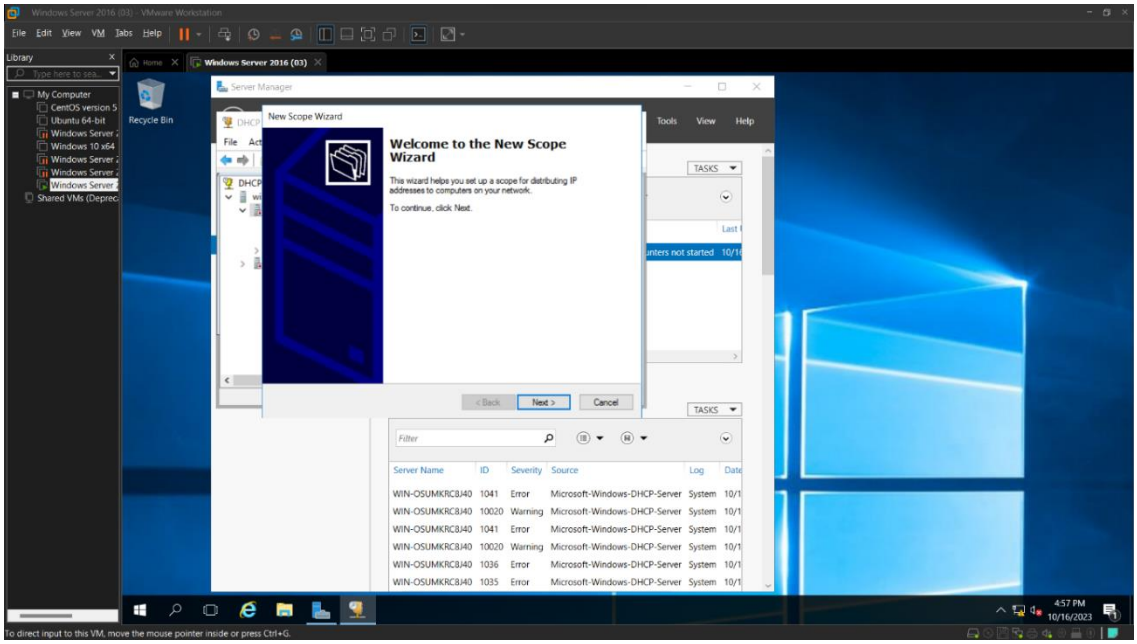


Figure 77 – Creating DHCP Scopes - Windows Server 2016

- IP Address Range Specifying

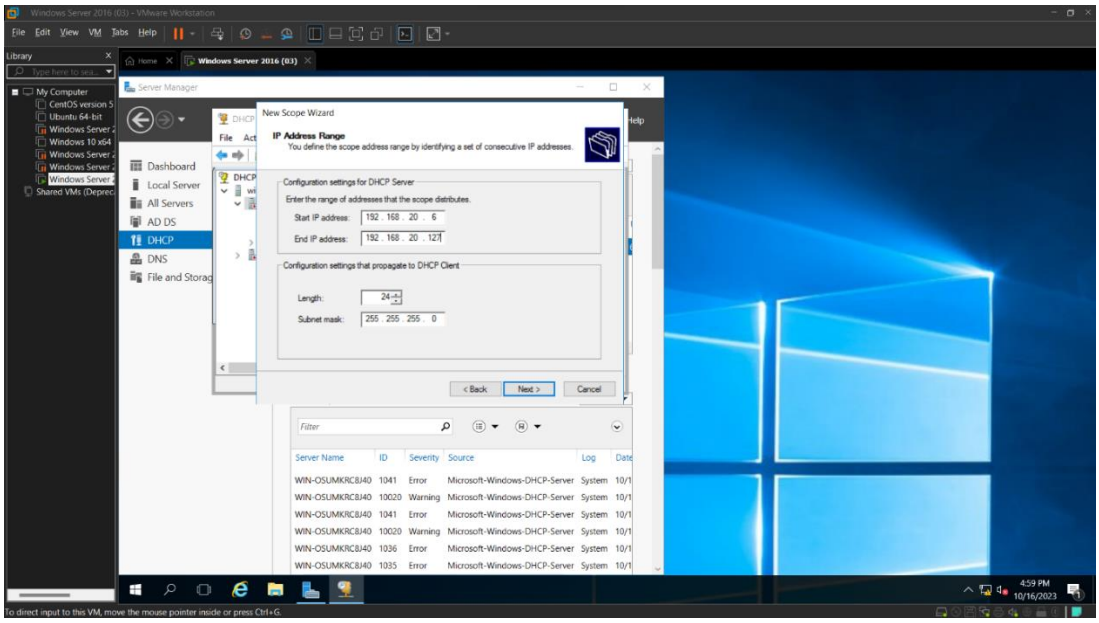


Figure 78 – IP Address Range Specifying - Windows Server 2016

- Adding Default Gateway

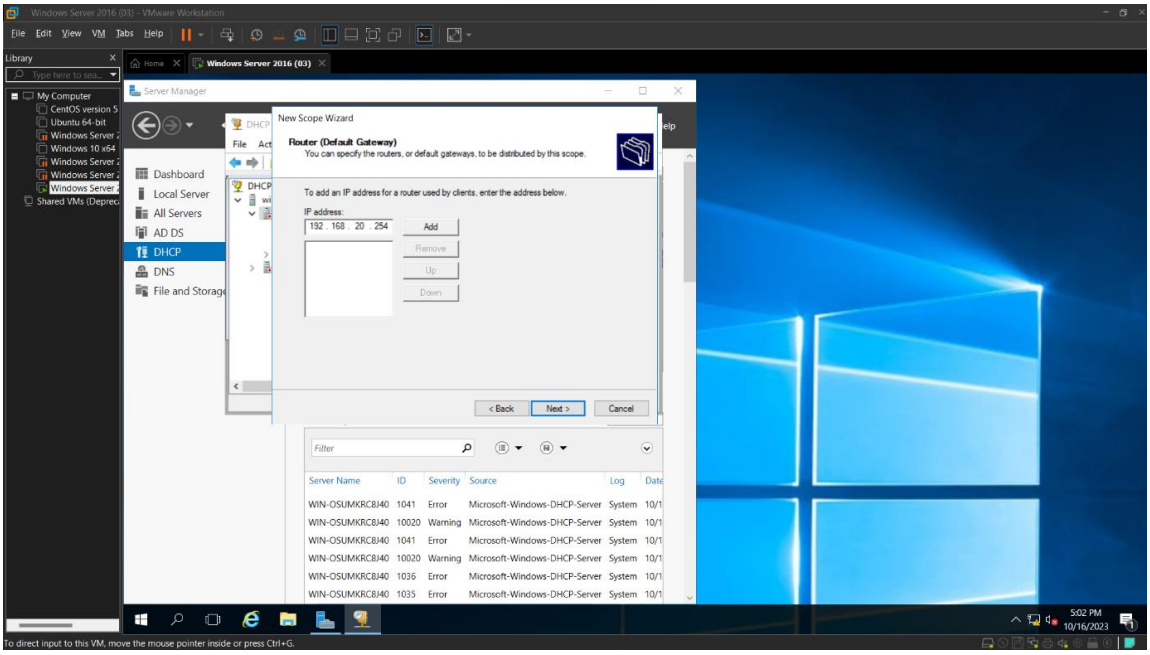


Figure 79 – Adding Default Gateway - Windows Server 2016

- Admin Pc Address Pool

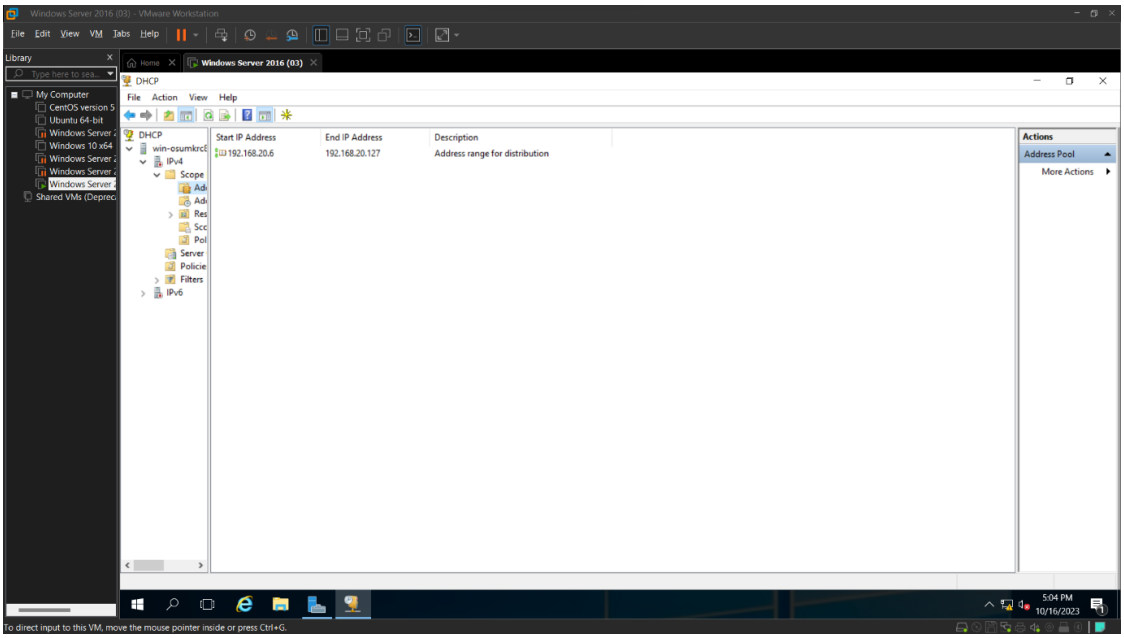


Figure 80 – Admin_Pc Address Pools - Windows Server 2016

- DHCP Pools

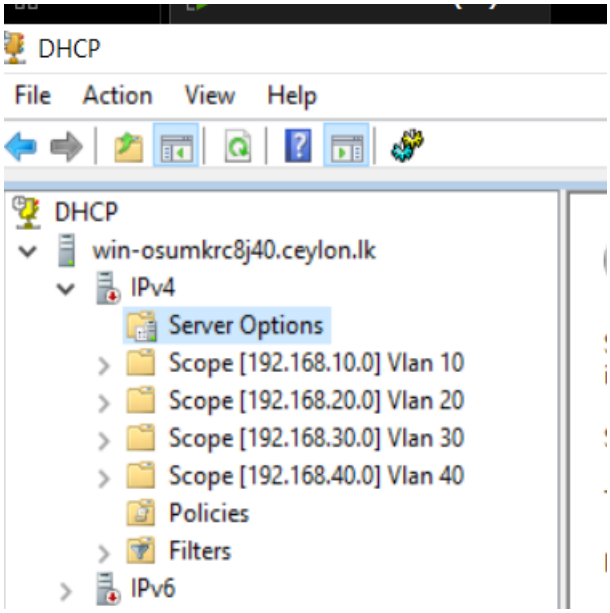


Figure 81 –DHCP Pools - Windows Server 2016

CHAPTER 5:

RESULTS AND EVALUVATION

5.1 IMPLEMENTATION VERIFICATION

To expedite the verification process, the output of server settings can be combined with Cisco switch show commands.

5.1.1 Core Layer switches Verification :

5.1.1.1 L3_Main Switch

- Show running config – Show the implementation's outcome

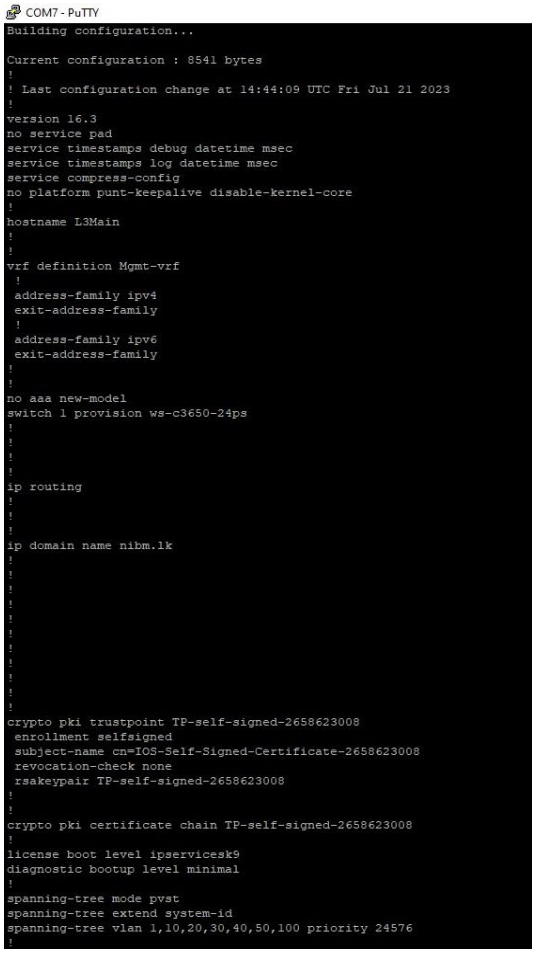


Figure 82 – Running configuration 1 – Layer 3 switch

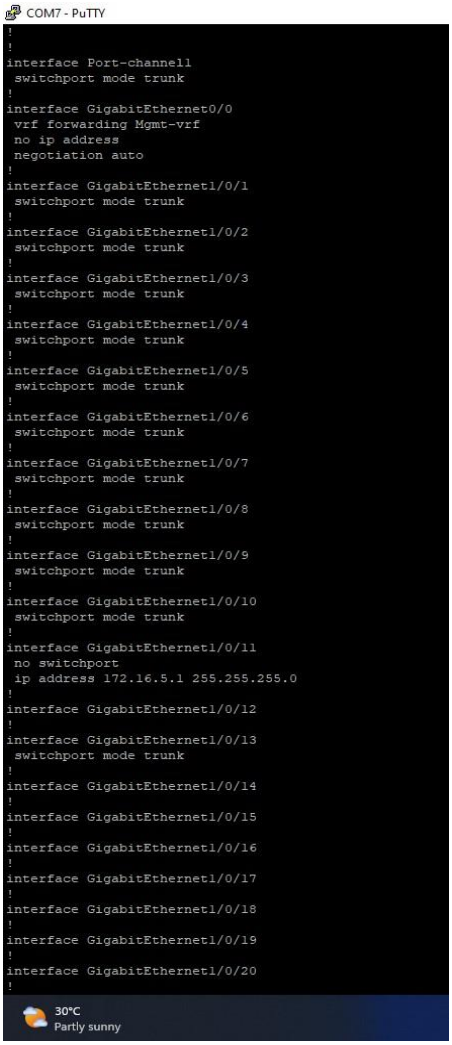


Figure 83 –Running config 2– Layer 3 switch

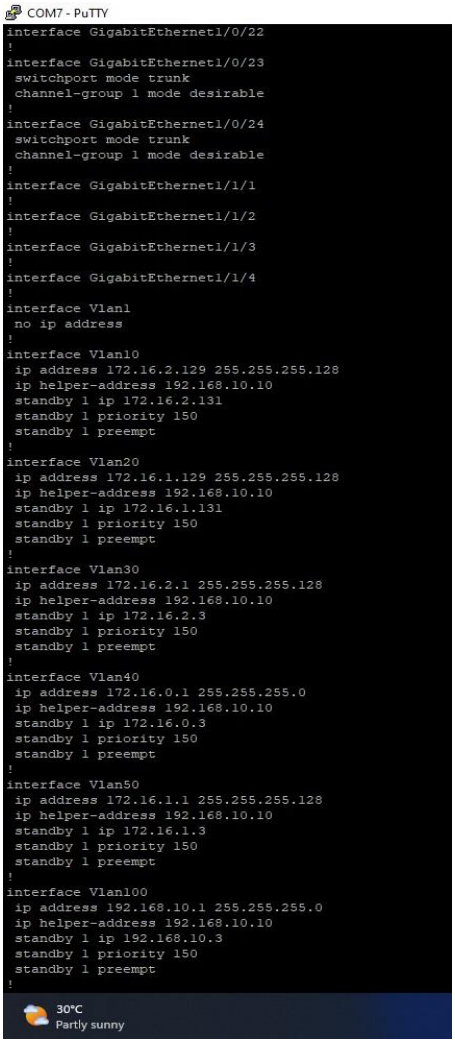


Figure 84 – Running config 3 - Layer 3 switch

- Show IP SSH :- Remote access verification



Figure 85 – SSH Verification– Layer 3 switch

- Show VLAN brief :- VLANs Verification

```
L3Main#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi1/0/1, Gi1/0/4, Gi1/0/5
                                                Gi1/0/6, Gi1/0/7, Gi1/0/8
                                                Gi1/0/9, Gi1/0/10, Gi1/0/12
                                                Gi1/0/13, Gi1/0/14, Gi1/0/15
                                                Gi1/0/16, Gi1/0/17, Gi1/0/18
                                                Gi1/0/19, Gi1/0/20, Gi1/0/21
                                                Gi1/0/22, Gi1/1/1, Gi1/1/2
                                                Gi1/1/3, Gi1/1/4
10   Marketing_Dept                   active
20   Finance_Dept                     active
30   Machanical_Dept                  active
40   It_Dept                          active
50   Hr_Dept                          active
100  Server                           active
101  Management_Vlan                  active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

Figure 86 – VlAN Verification – Layer 3 switch

- Show VTP status :- VTP Verification

```
L3Main(config)#
L3Main(config)#
L3Main(config)#do sh vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : nibm.lk
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0007.EC15.4200
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
--------------
VTP Operating Mode                : Server
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 5
Configuration Revision            : 0
MD5 digest                        : 0xFE 0xF8 0x77 0x74 0xBD 0xC7 0x1E 0x0D
                                    0x86 0xC5 0xE2 0x84 0x70 0x84 0xC0 0x1D

L3Main(config)#
L3Main(config)#
```

Figure 87 – VTP Verification – Layer 3 switch

- Show standby brief :- HSRP Verification

```
L3Main#show stan
L3Main#show standby br
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
Vl10        1    150 P Active  local           172.16.2.130    172.16.2.131
Vl20        1    150 P Active  local           172.16.1.130    172.16.1.131
Vl30        1    150 P Active  local           172.16.2.2      172.16.2.3
Vl40        1    150 P Active  local           172.16.0.2      172.16.0.3
Vl50        1    150 P Active  local           172.16.1.2      172.16.1.3
Vl100       1    150 P Active  local           192.168.10.2    192.168.10.3
L3Main#
```

Figure 88 – HSRP Verification – Layer 3 switch

- Show EtherChannel summary :- Ether channel Verification

```
L3_Main#sh etherchannel
                Channel-group listing:
                ----------------------

Group: 1
----------
Group state = L2
Ports: 2   Maxports = 8
Port-channels: 1 Max Port-channels = 1
Protocol:   PAgP
Minimum Links: 0



L3_Main#
```

Figure 89– ETHER-CHANNEL Verification – Layer 3 switch

- Assigning IP Address to VLANs Verification

```
!
interface Vlan10
 ip address 172.16.2.129 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.2.131
 standby 1 priority 150
 standby 1 preempt
!
interface Vlan20
 ip address 172.16.1.129 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.1.131
 standby 1 priority 150
 standby 1 preempt
!
interface Vlan30
 ip address 172.16.2.1 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.2.3
 standby 1 priority 150
 standby 1 preempt
!
interface Vlan40
 ip address 172.16.0.1 255.255.255.0
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.0.3
 standby 1 priority 150
 standby 1 preempt
!
interface Vlan50
 ip address 172.16.1.1 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.1.3
 standby 1 priority 150
 standby 1 preempt
!
interface Vlan100
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.10.10
 standby 1 ip 192.168.10.3
 standby 1 priority 150
 standby 1 preempt
!
```

Figure 90 – IP Assign for VlAN Interface Verification – Layer 3 switch

- Show IP route :- IP Routing Verification

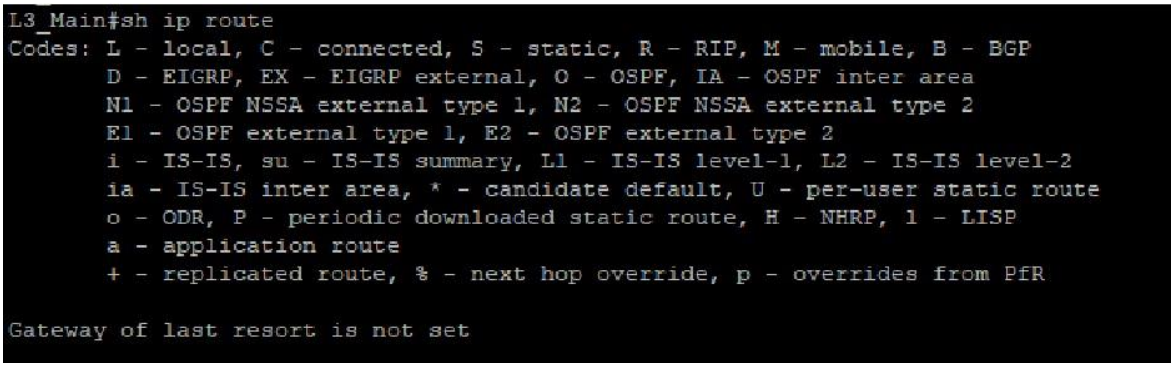


Figure 91 –Routing Verification – Layer 3 switch

5.1.1.2 L3_Backup switch

Layer 3 switch has been used here as a backup device of the main Layer 3 switches. Once the main path is down, backup path has been directed via this device.

- Show running-config :- Show the implementation's outcome

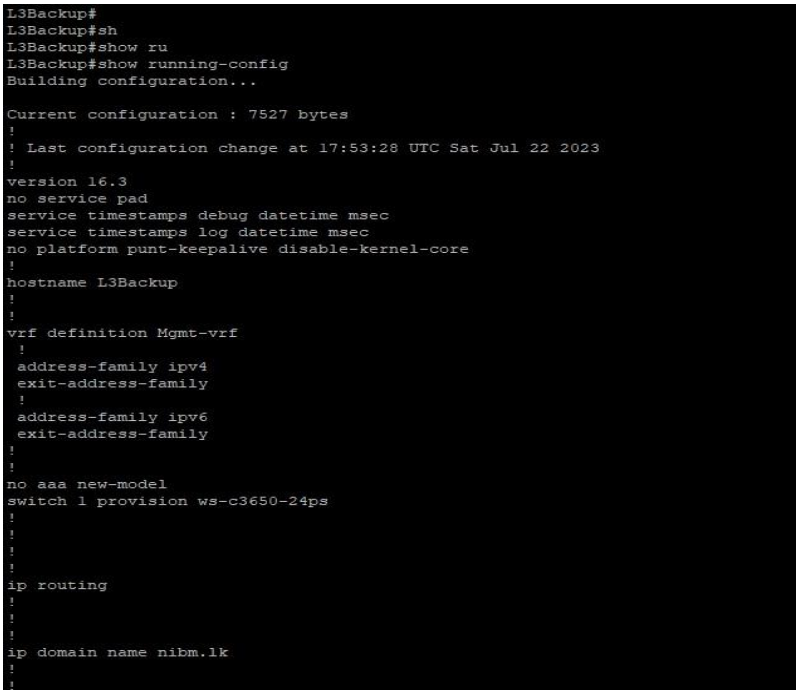


Figure 92 – Result 1 – Layer 3 Backup switch

COM7 - PuTTY

```
!
!
interface Port-channel1
 switchport mode trunk
!
interface GigabitEthernet0/0
 vrf forwarding Mgmt-vrf
 no ip address
 negotiation auto
!
interface GigabitEthernet1/0/1
!
interface GigabitEthernet1/0/2
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
 no switchport
 ip address 172.16.5.2 255.255.255.0
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
!
interface GigabitEthernet1/0/15
!
interface GigabitEthernet1/0/16
!
interface GigabitEthernet1/0/17
!
interface GigabitEthernet1/0/18
!
interface GigabitEthernet1/0/19
!
interface GigabitEthernet1/0/20
!
interface GigabitEthernet1/0/21
!
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
 switchport mode trunk
 channel-group 1 mode desirable
!
interface GigabitEthernet1/0/24
 switchport mode trunk
 channel-group 1 mode desirable
```

Figure 93 – Result 2 – Layer 3 Backup switch

COM7 - PuTTY

```
interface GigabitEthernet1/0/22
!
interface GigabitEthernet1/0/23
 switchport mode trunk
 channel-group 1 mode desirable
!
interface GigabitEthernet1/0/24
 switchport mode trunk
 channel-group 1 mode desirable
!
interface GigabitEthernet1/1/1
!
interface GigabitEthernet1/1/2
!
interface GigabitEthernet1/1/3
!
interface GigabitEthernet1/1/4
!
interface Vlan1
 no ip address
!
interface Vlan10
 ip address 172.16.2.130 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.2.131
!
interface Vlan20
 ip address 172.16.1.130 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.1.131
 standby 1 preempt
!
interface Vlan30
 ip address 172.16.2.2 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.2.3
 standby 1 preempt
!
interface Vlan40
 ip address 172.16.0.2 255.255.255.0
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.0.3
!
interface Vlan50
 ip address 172.16.1.2 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.1.3
 standby 1 preempt
!
interface Vlan100
 ip address 192.168.10.2 255.255.255.0
 ip helper-address 192.168.10.10
 standby 1 ip 192.168.10.3
 standby 1 preempt
!
interface Vlan101
 ip address 192.168.11.2 255.255.255.0
!
ip forward-protocol nd
ip http server
ip http secure-server
ip route 0.0.0.0 0.0.0.0 172.16.5.3
!
```

Figure 94– Result 3– Layer 3 Backup switch

- Show VLAN:- VLANs Verification

```
L3Backup#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi1/0/2, Gi1/0/4, Gi1/0/5
                                                Gi1/0/6, Gi1/0/7, Gi1/0/8
                                                Gi1/0/9, Gi1/0/10, Gi1/0/11
                                                Gi1/0/13, Gi1/0/14, Gi1/0/15
                                                Gi1/0/16, Gi1/0/17, Gi1/0/18
                                                Gi1/0/19, Gi1/0/20, Gi1/0/21
                                                Gi1/0/22, Gi1/1/1, Gi1/1/2
                                                Gi1/1/3, Gi1/1/4
10   Marketing_Dept                   active
20   Finance_Dept                     active
30   Machanical_Dept                  active
40   It_Dept                          active
50   Hr_Dept                          active
100  Server                           active
101  Management_Vlan                  active
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

Figure 95 – VLAN Verification – Layer 3 Backup switch

- Show standby brief :- HSRP Verification

```
L3Backup#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State   Active          Standby         Virtual IP
V110        1    100   Standby 172.16.2.129    local           172.16.2.131
V120        1    100 P Standby 172.16.1.129    local           172.16.1.131
V130        1    100 P Standby 172.16.2.1      local           172.16.2.3
V140        1    100   Standby 172.16.0.1      local           172.16.0.3
V150        1    100 P Standby 172.16.1.1      local           172.16.1.3
V1100       1    100 P Standby 192.168.10.1    local           192.168.10.3
L3Backup#
```

Figure 96 – HSRP Verification – Layes 3 Backup Switch

- Assigning IP Address to VLANs Verification

```
interface Vlan10
 ip address 172.16.2.130 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.2.131
!
interface Vlan20
 ip address 172.16.1.130 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.1.131
 standby 1 preempt
!
interface Vlan30
 ip address 172.16.2.2 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.2.3
 standby 1 preempt
!
interface Vlan40
 ip address 172.16.0.2 255.255.255.0
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.0.3
!
interface Vlan50
 ip address 172.16.1.2 255.255.255.128
 ip helper-address 192.168.10.10
 standby 1 ip 172.16.1.3
 standby 1 preempt
!
interface Vlan100
 ip address 192.168.10.2 255.255.255.0
 ip helper-address 192.168.10.10
 standby 1 ip 192.168.10.3
 standby 1 preempt
!
interface Vlan101
 ip address 192.168.11.2 255.255.255.0
!
ip forward-protocol nd
ip http server
ip http secure-server
ip route 0.0.0.0 0.0.0.0 172.16.5.3
!
```

Figure 97– IP Assign for VlAN Interface Verification – Layer 3 Backup switch Show VTP status :- VTP Verification

- VTP Verification

```
L3Backup(config)#
L3Backup(config)#do sh vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : nibm.lk
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0007.EC15.4200
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00
Local updater ID is 0.0.0.0 (no valid interface found)

Feature VLAN :
--------------
VTP Operating Mode              : Server
Maximum VLANs supported locally : 1005
Number of existing VLANs        : 5
Configuration Revision          : 0
MD5 digest                      : 0xFE 0xF8 0x77 0x74 0xBD 0xC7 0x1E 0x0D
```

Figure 98 – VTP Verification – Layer 3 Backup switch

5.1.2 Layer 2 Devices Verifications

5.1.2.1 Switch Verification : SW_01/ SW_02

The Access layer of the network has been made easily accessible to users. To ensure security, every access layer switch has been configured with SSH and strong passwords. Port security measures have also been implemented to enhance network security. The performance of network devices is regulated through the use of VLANs**.**

- VTP Verification (SW_01)

```
SW_01#
SW_01#sh vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : nibm.lk
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 0010.115A.5900
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00

Feature VLAN :
--------------
VTP Operating Mode              : Client
Maximum VLANs supported locally : 255
Number of existing VLANs        : 5
Configuration Revision          : 0
MD5 digest                      : 0xFE 0xF8 0x77 0x74 0xBD 0xC7 0x1E 0x0D
                                  0x86 0xC5 0xE2 0x84 0x70 0x84 0xC0 0x1D

SW_01#
SW_01#
```

Figure 99– VTP Verification – Switch 01

- Truncking ports Verification

```
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport mode trunk
!
interface FastEthernet0/3
 switchport mode trunk
!
```

Figure 100– Port Verification – Switch 01

- VTP Verification (SW_02)

```
SW_02(config)#do sh vtp status
VTP Version capable             : 1 to 2
VTP version running             : 1
VTP Domain Name                 : nibm.lk
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : 00D0.979C.A300
Configuration last modified by 0.0.0.0 at 0-0-00 00:00:00

Feature VLAN :
--------------
VTP Operating Mode                : Client
Maximum VLANs supported locally   : 255
Number of existing VLANs          : 5
Configuration Revision            : 0
MD5 digest                        : 0xFE 0xF8 0x77 0x74 0xBD 0xC7 0x1E 0x0D
                                    0x86 0xC5 0xE2 0x84 0x70 0x84 0xC0 0x1D
```

Figure 101– VTP Verification – Switch 02

- VLAN Access Verification (SW_01)

```
interface FastEthernet0/4
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 20
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 20
 switchport mode access
!
```

Figure 102– VLAN Access Verification – Switch 01

- VLAN Access Verification (SW_02)

```
interface FastEthernet0/4
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/6
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/7
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/8
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/9
 switchport access vlan 40
 switchport mode access
```

Figure 103– VLAN Access Verification – Switch 02

- Port Security Verification – show port-security (command)

```
interface FastEthernet0/4
 switchport access vlan 30
 switchport mode access
 switchport port-security maximum 5
 switchport port-security mac-address sticky
 switchport port-security violation restrict
!
```

Figure 104– Port Security Verification – Switch 02

5.1.3 Router Verification

SHH Verification

```
SSH Enabled - version 1.99
Authentication methods:publickey,keyboard-interactive,password
Encryption Algorithms:aes128-ctr,aes192-ctr,aes256-ctr,aes128-cbc,3des-cbc,aes192-cbc,aes256-cbc
MAC Algorithms:hmac-sha1,hmac-sha1-96
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAAAgQCcGEFsBYAZd754XlVJqk4DWQWzf7qT+DpVXfBKD4Ut
4EMJPUGtJODxztUvipjybypvVFVlKfXYkI9ECVJVw+zxRgO4lLaNcLCiE0JmPQYRuPG3Y+/kSpK6C3Db
je8/Q2Z/SZKblqKvabjEzA4MTX55+xLd3R3R2utXfToyTDBCUQ==
```

Figure 105– SSH Verification – Router

- Routing Verification

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is 192.168.16.254 to network 0.0.0.0

S*    0.0.0.0/0 [254/0] via 192.168.16.254
      1.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        1.1.1.0/29 is directly connected, GigabitEthernet0/0
L        1.1.1.1/32 is directly connected, GigabitEthernet0/0
      192.168.2.0/32 is subnetted, 1 subnets
S        192.168.2.6 [254/0] via 192.168.16.254, GigabitEthernet0/1
      192.168.16.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.16.0/24 is directly connected, GigabitEthernet0/1
L        192.168.16.28/32 is directly connected, GigabitEthernet0/1
D     192.168.20.0/24 [90/28416] via 1.1.1.3, 00:08:41, GigabitEthernet0/0
D     192.168.30.0/24 [90/28416] via 1.1.1.3, 00:08:41, GigabitEthernet0/0
D     192.168.40.0/24 [90/28416] via 1.1.1.3, 00:08:41, GigabitEthernet0/0
D     192.168.50.0/24 [90/28416] via 1.1.1.3, 00:08:41, GigabitEthernet0/0
```

Figure 106 – Routing Verification – Routers

5.2 ACHIEVED RESULTS

When the network configurations are complete, the results of such implementations can be seen. These results show that the network's design and implementation were both successfully carried out.

- Ping Verification

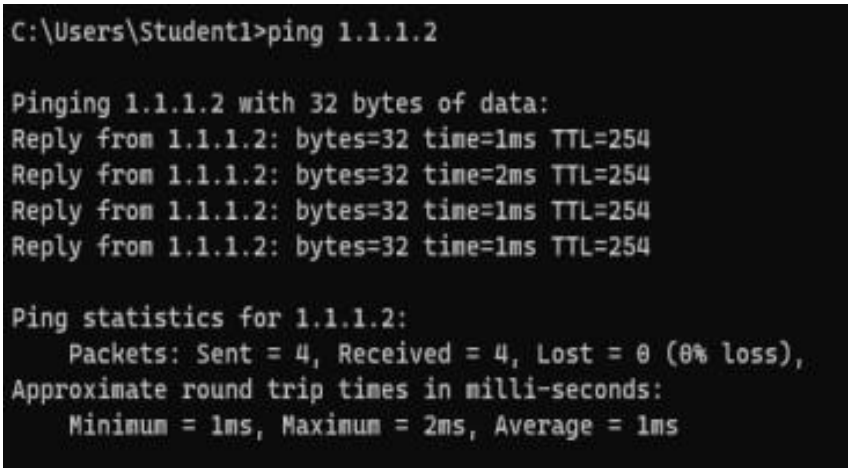


Figure 107 – Ping Verification

- Internet connection Verification

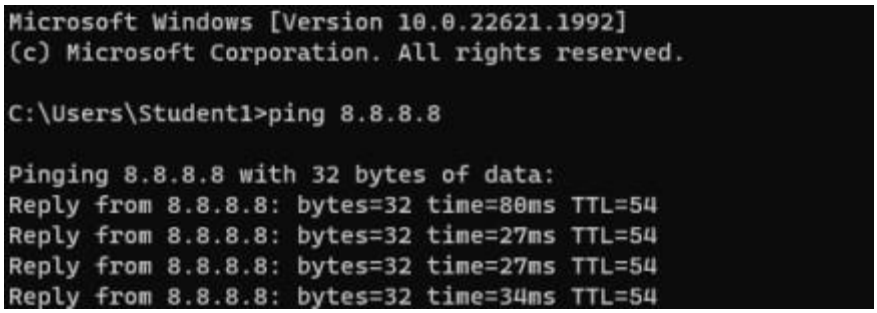


Figure 108– Internet Connection Verification – Router

Chapter 6:

CONCLUTION

In the end, our network company serves as a prime illustration of the harmonious coexistence of advanced technology and an unwavering dedication to excellence. Through our three-story headquarters, we have not only optimized space utilization but also fostered an environment that promotes effective teamwork among different departments.

Our dedication to stringent security procedures is essential in safeguarding sensitive data and invaluable assets from external threats. By providing our clients with the reassurance they require, our robust security framework, featuring state-of-the-art access controls and surveillance technologies, acts as a formidable defense against any potential breaches.

Moreover, our comprehensive cabling infrastructure acts as the cornerstone for efficient data transmission and communication, ensuring a continuous exchange of information throughout the company. This not only positions us at the forefront of technological advancement but also bolsters our capacity for future expansion and enhances operational effectiveness.

As we set our sights on the future, our network company remains unwavering in its commitment to maintaining state-of-the-art security. We will continue to allocate resources to research and development, ensuring our reliability as a trusted partner for enterprises seeking network solutions.

Put plainly, our network company with three floors symbolizes our unwavering commitment to excellence and our vision for a future where connectivity and security take center stage. We are eager to build and protect upcoming networks, facilitating the success of businesses in an interconnected world.

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ANNEXUER 1

DEVICE SPECIFICATION

- Access Point



Name	C9130AXI-H
Model	C9130AXI-H- Cisco Catalyst 9130 WIFI Access Point
Details	Cisco Catalyst 9130AX Series
Warranty	3 Years

More Details:

Description	Cisco Catalyst C9130AX Series, H Domain
Software	• Cisco Unified Wireless Network Software Release 8.10.x or later • Cisco IOS ® XE Software Release 16.12.1 with AP Device Pack, or later

Supported Wireless LAN Controllers	• Cisco Catalyst 9800 Series Wireless Controllers • Cisco 3504, 5520, and 8540 Wireless Controllers and Cisco Virtual Wireless Controller

802.11n version 2.0 (and related) capabilities	• 4x4 MIMO with four spatial streams • Maximal Ratio Combining (MRC) • 802.11n and 802.11a/g • 20- and 40-MHz channels • PHY data rates up to 1.5 Gbps (40 MHz with 5 GHz) • Packet aggregation A-MPDU (transmit and receive), A-MSDU) (transmit and receive) • 802.11 DFS • CSD support
802.11ac	• 8x8 downlink MU-MIMO with eight spatial streams • MRC • 802.11ac beamforming • 20-, 40-, 80-, and 160-MHz channels • PHY data rates up to 6.9 Gbps (160 MHz with 5 GHz) • Packet aggregation: A-MPDU (transmit and receive), A-MSDU (transmit and receive) • 802.11 DFS • CSD support • WPA3 support

802.11ax	• 8x8 uplink/downlink MU-MIMO with eight spatial streams • Uplink/downlink OFDMA • TWT • BSS coloring • MRC • 802.11ax beamforming • 20-, 40-, 80-, and 160-MHz channels • PHY data rates up to 10 Gbps (160 MHz with 5 GHz and 20 MHz with 2.4 GHz) • Packet aggregation: A-MPDU (transmit and receive), A-MSDU (transmit and receive) • 802.11 DFS • CSD support • WPA3 support
Integrated antenna	• 2.4 GHz: Peak gain 4 dBi, internal antenna, omnidirectional in azimuth • 5 GHz: Peak gain 6 dBi, internal antenna, omnidirectional in azimuth
External antenna (sold separately)	• The Cisco Catalyst 9130AXE Access Point is certified for use with antenna gains up to 13 dBi (2.4 GHz and 5 GHz) • Cisco offers the industry's broadest selection of antennas, delivering optimal coverage for a variety of deployment scenarios
Interfaces	• 1x 100, 1000, 2500, 5000 Multigigabit Ethernet (RJ-45) – IEEE 802.3bz • Management console port (RJ-45) • USB 2.0 at 4.5W (enabled via future software
Indicators	• Status LED indicates boot loader status, association status, operating status, boot loader warnings, and boot loader errors
Dimensions(W x L x H)	• Access point (without mounting brackets): - C9130AXI: 8.9 x 8.9 x 1.88 in. (22.6 x 22.6 x 4.8 cm)

Environmental	Cisco Catalyst 9130AXI • Nonoperating (storage) temperature: - 22° to 158°F (-30° to 70°C) • Nonoperating (storage) altitude test: 25˚C, 15,000 ft (4600 m) • Operating temperature: 32° to 122°F (0° to 50°C) • Operating humidity: 10% to 90% (noncondensing) • Operating altitude test: 40˚C, 9843 ft (3000 m) Note: When the ambient operating temperature exceeds 40°C, the access point will shift from 8x8 to 4x4 on the 5 GHz radio, uplink Ethernet will downgrade to 1 Gigabit Ethernet; however, the USB interface will remain enabled
System Memory	• 2048 MB DRAM • 1024 MB flash

- Server Rack



Model : Cisco R42610 Rack

Cisco R42610 Rack	Standard	Expansion
Dimensions (H x W x D)	78.74 x 24 x 43.38 in. (/2000 x 610 x 1102 mm)	78.74 x 23.58 x 43.38 in. (/2000 x 599 x 1102 mm)
Side panels included	Yes	No
Equipment mounting capacity	42RU	42RU
Static load capacity	2100 lb (/954 kg)	2100 lb (/954 kg)
Dynamic load capacity	Not applicable	Not applicable

- Server



Name	Rack Server
Model	PowerEdge R760
Warranty	3 Years

More Details:

Processor	Up to two 4th Generation Intel® Xeon® Scalable processors
Memory	Up to 32 x DDR5 RDIMMs
Workloads	High Performance Simulation & Modeling, OLTP and All-Flash SDS
Storage Controllers	• Internal Controllers: PERC H965i, PERC H755, PERC H755N, PERC H355, HBA355i • Internal Boot: Boot Optimized Storage Subsystem (BOSS-N1): HWRAID 2 x M.2 NVMe SSDs or USB • External HBA (non-RAID): HBA355e • Software RAID: S160
Accelerators	Supports 3 x Single-Wide* GPUs
Power Supplies	• 2800 W Titanium • 2400 W Platinum • 1800 W Titanium • 1400 W Platinum • 1100 W Titanium • 1100 W LVDC -48 — -60 VDC, hot swap redundant • 800 W Platinum 100—240 VAC or 240 HVDC, hot swap redundant • 700 W Titanium 200—240 VAC or 240 HVDC, hot swap redundant

Dimensions	• Height – 86.8 mm (3.41 inches) • Width – 482 mm (18.97 inches) • Depth – 772.13 mm (30.39 inches) with bezel 758.29 mm (29.85 inches) without bezel

- Cisco Catalyst 3900L- 24P – 4X - E



Name	C9300L- 24P – 4X - E
Model	C9300L- 24P – 4X - E -Cisco Catalyst 9300L Switches
Details	Catalyst 9300 24-port fixed uplinks PoE+, 4X10G uplinks, Network Essentials
Warranty	3 Years
Power	715W
Uplink interface	24x 1G fixed Uplinks
Ports	24 (PoE+ compatible)

- Router



Name	C8300 –2N2S - 4T2X
Model	C8300-2N2S-4T2X - Cisco Catalyst 8300 Series Edge Platforms Series
Details	C8300 2RU w/ 10G WAN (2 SM and 2 NIM slots, and 2 x 10-Gigabit Ethernet and 4 x 1- Gigabit Ethernet ports
Warranty	3 Years

More Details :

Flash memory	8 GB (can be upgraded)
DRAM	8 GB (can be upgraded)
SD – WAN IPsec throughput	Up to 18 Gbps
IPv4 forwarding throughput	Up to 19.7 Gbps
Storage	16G M.2 (can be upgraded)
Power	Dual power supply
Modularity and form factor	1RU and 2RU form factor Supports SM, NIM, and Pluggable Interface Module (PIM) slots

- Power Supply



Name	PWR-C2-1025WAC=
Model	PWR-C2-1025WAC= Catalyst 3650 Series Spare Power Supply
Warranty	3 Years

More Details :

Max. Powe Supply Rate	1025W
Total Output BTU (Note: 1000 BTU per hr = 293 W)	3497 BTU/hr, 1025W
Input-Voltage Range and Frequency (Note: ±10% of "Input-Voltage Range")	115-240VAC, 50-60 Hz
Input Current	12-6A
Output Ratings	12V @ 20.83A -54V @ 14.6A
Output Holdup Time	10 ms minimum @ 102.5VAC
Power-Supply Input Receptacles	IEC 320-C16 (IEC60320-C16)

Power Cord Rating	13A

Physical Specifications	(H x W x D): 1.58 X 3.75 X 13.25 in (4.0 x 9.5 x 33.7 cm) Weight: 3.55 lb (1.61 kg)

- IP Cameras



Model – 4MP IR Bullet Network Camera Name - DS-2CD2T45G0P-I

- High quality imaging with 4 MP resolution
- 180°super wide field of view
- Efficient H.265+ compression technology
- Clear imaging against strong back light due to 120 dB true WDR technology
- Distortion correction
- Water and dust resistant (IP67)

Sp**ecifications:**

Camera	
Image Sensor	1/2.7" Progressive Scan CMOS
Min. Illumination	0.028 Lux @ (F2.0, AGC ON), 0 Lux with IR
WDR	120 dB
Angle Adjustment	Pan: 0° to 360°, tilt: 0° to 90°, rotate: 0° to 360°
Day & Night	ICR cut filter
Shutter Speed	1/3 s to 1/100,000 s

Lense	
Lens Type	1.68mm
Focus	Fixed
Aperture	F2.0
FOV	Horizontal: 180°, vertical: 101°, diagonal: 180°
Lens Mount	M12

Video	
Max. Resolution	2688 x 1520
Main Stream	50Hz: 25fps (2688 × 1520, 2304 × 1296, 1920 × 1080, 1280 × 720) 60Hz: 24fps (2688 × 1520), 30fps (2304 × 1296, 1920 × 1080, 1280 × 720)
Sub-Stream	50Hz: 25fps (640 × 480, 640 × 360, 320 × 240) 60Hz: 30fps (640 × 480, 640 × 360, 320 × 240)
Video Compression	Main stream: H.265+/H.265/H.264+/H.264 Sub stream: H.265/H.264 /MJPEG
Illuminator	
IR Range	20 m
Wavelength	850 nm
Smart Supplement Light	Yes
Image	

Image Enhancement	BLC,HLC,EIS,Lens distortion,3D DNR
Image Setting	Saturation, brightness, contrast, Sharpness, Rotate mode, White balance, AGC
Day & Night Switch	Day/Night/Auto/Schedule
SNR	>=50 DB

Network	
Network Storage	Micro SD/SDHC /SDXC (256 G) local storage, and NAS (NFS, SMB/CIFS), auto network replenishment (ANR)
Protocols	TCP/IP, ICMP, HTTP, HTTPS, FTP, DHCP, DNS, DDNS, RTP, RTSP, PPPoE, NTP, UPnP, SMTP, SNMP, IGMP, 802.1X, QoS, IPv6, UDP, Bonjour, SSL/TLS
API	Password protection, complicated password, HTTPS encryption, 802.1X authentication (EAPTLS 1.2, EAP-LEAP, EAP-MD5)
User/Host	Up to 32 users, 3 levels: administrator, operator and user
Client	iVMS-4200, Hik-Connect, Hik-Central
Web Browser	Plug-in required live view: Chrome 45.0+, Firefox 52.0+ Plug-in free live view: Chrome 41.0+, Firefox 30.0+